GEOGRAPHY ESSENTIALS

Understanding diagrams

Susan Watson

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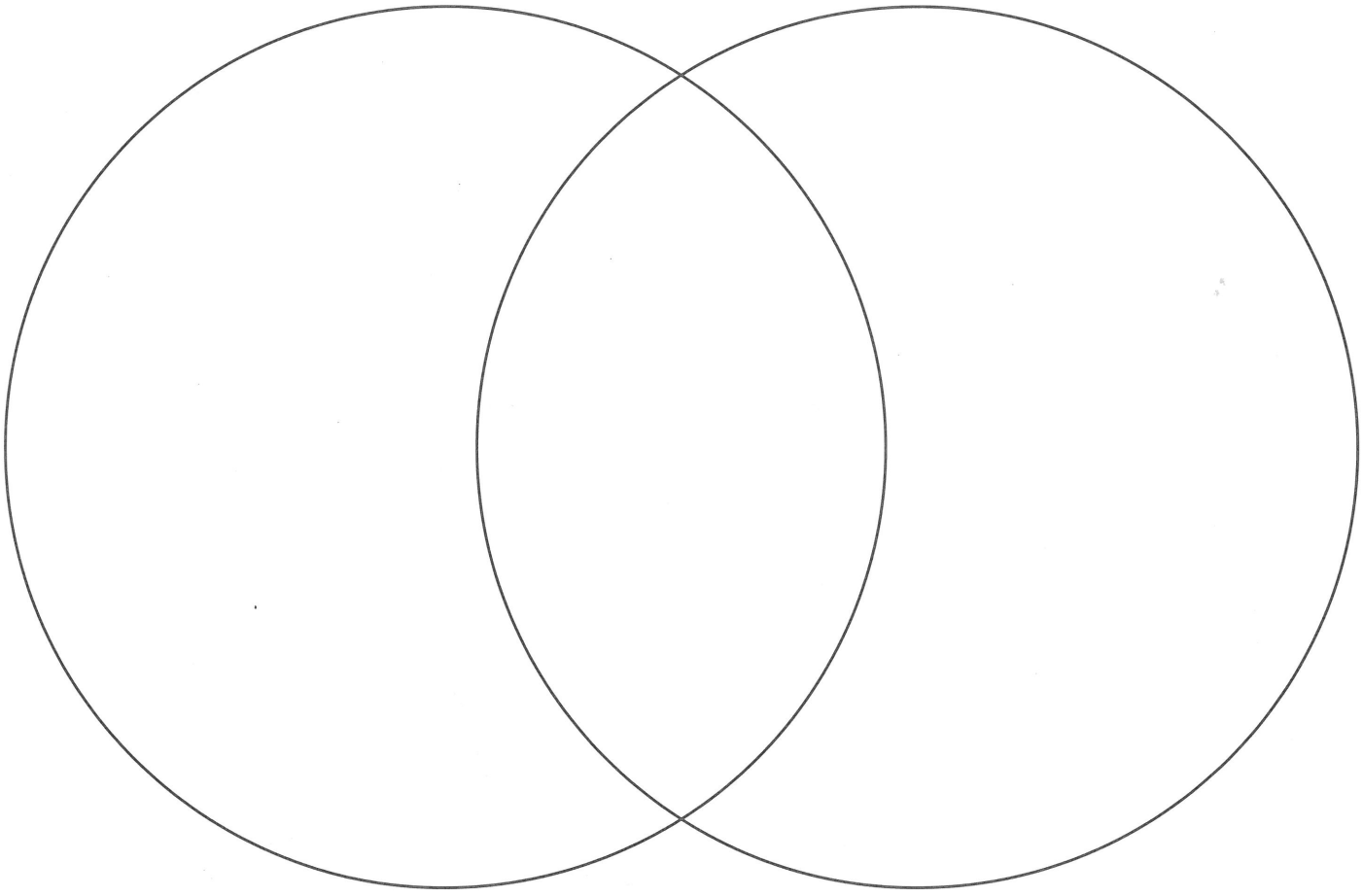

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SOLUTION AND REASON: _____

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GEOGRAPHY ESSENTIALS

Understanding diagrams

Susan Watson

The author, Susan Watson, combines her teaching and publishing backgrounds in the writing of a wide range of educational materials, particularly in the field of Geography.

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Introduction

The **Geography Essentials** series provides photocopiable worksheets that reinforce essential geographic skills and knowledge. They could be used with students from junior to middle secondary levels.

Understanding Diagrams explores geographic concepts through the use of a range of different types of diagrams – schematic, block, cross-section, profile, transect, flow, concept and organisation. Topics cover both physical and human aspects of Geography, and are based on the knowledge students should have at years 7–10. Students not only interpret diagrams but also complete outlines and build up their own.

Each topic covers two pages and most provide space for students to answer directly onto them. As the diagrams are not in colour, it would be a valuable additional activity to ask students to colour them if not already part of the worksheet tasks. Using the appropriate colours on diagrams is another helpful learning technique.

Each worksheet also contains a short introductory statement and a glossary of geographic terms central to the topic. Teachers are encouraged to discuss these concepts as a way of preparing students for the activities to be completed on the worksheet.

The worksheets should not be seen as complete 'lessons' in themselves. They are not a replacement for a textbook and/or teacher instruction. Rather they provide interesting activities that are sometimes a little outside the usual textbook square.

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1 Diagrams in Geography

When you study Geography, you learn about earth's natural features – land, water, and plant and animal life. You also look at places and things people have built and try to understand how and why they are different, and what effect they have had on natural things. You've probably heard the saying: 'A picture's worth a thousand words'. Pictures in Geography are called diagrams. Diagrams in Geography are graphic organisers that help you understand more about Planet Earth and how people live on it and interact with it.

DEFINITIONS

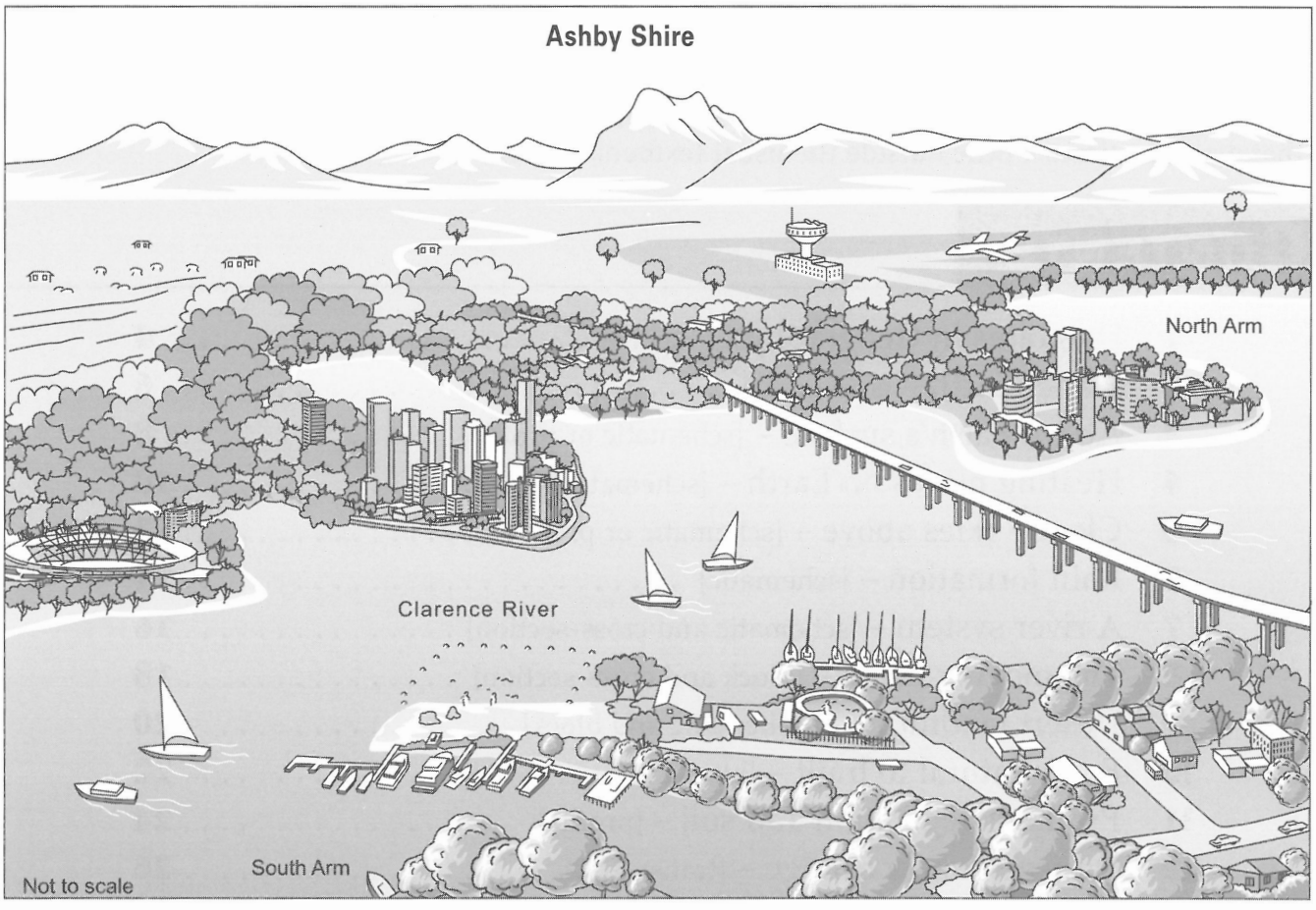
Geography is the study of Earth as the place where people live

diagrams are special types of drawings that show models of things in real life, or how things work, or steps in a process, or how things are organised

to **interact** is when two or more things act on each other by coming into contact

the **built environment** is what people have made or grown

the **natural environment** is all the living and non-living parts of Earth that have been formed by nature



1 The diagram above shows a small part of Earth's surface where the built and natural environments mix, and includes some human activity.

a List 4 things in the built environment.

_____

_____

_____

_____

b List 4 things in the natural environment.

_____

_____

_____

_____

c List 3 different activities you can see people doing.

_____ _____ _____

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d Write your own description of what the diagram is telling you.

2 Swap your description with a classmate and read each other's. Discuss between yourselves the advantages and/or disadvantages of the diagram compared to the written description, and then complete this table:

	Advantage	Disadvantage
Diagram		
Written description		

3 Add these labels to the diagram and 4 others of your own:

snowfields sports ground marina Centenary Bridge

4 Explain what the statement at the bottom of the diagram means. Why is it shown?

5 What do you think is the meaning of the saying: 'A picture's worth a thousand words'?

6 Adding colour to diagrams makes them easier to understand. Colour the diagram appropriately.

7 Look at the different sorts of diagrams in a Geography textbook and complete this table.

Type of diagram	Page number	Title or description of diagram
Built environment		1
		2
Natural environment		1
		2
Combination		1
		2

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2 Planet Earth

All living and non-living things on Earth are part of a large planet in one of the solar systems of the universe. Earth is the third planet from the Sun at a distance of about 150 million kilometres. Earth is the fifth largest planet out of the nine known planets in our solar system. Earth's diameter is about 12 750 kilometres, compared to the Sun's diameter of 1 392 000 kilometres. Earth has one moon, which is about one-quarter the size of Earth.

DEFINITIONS

a **planet** is a large spherical mass revolving in space around the sun

diameter is the straight-line distance from one side of a circle through the centre to the opposite side

molten rock has become liquid due to extremely high temperature and pressure inside Earth

Earth in our solar system

1 Name the nine known planets in our solar system in order from the Sun. The first letters have been given to help you. Start on the top row and move across to the right, then the second and third rows.

M_____ V_____ E_____

M_____ J_____ S_____

U_____ N_____ P_____

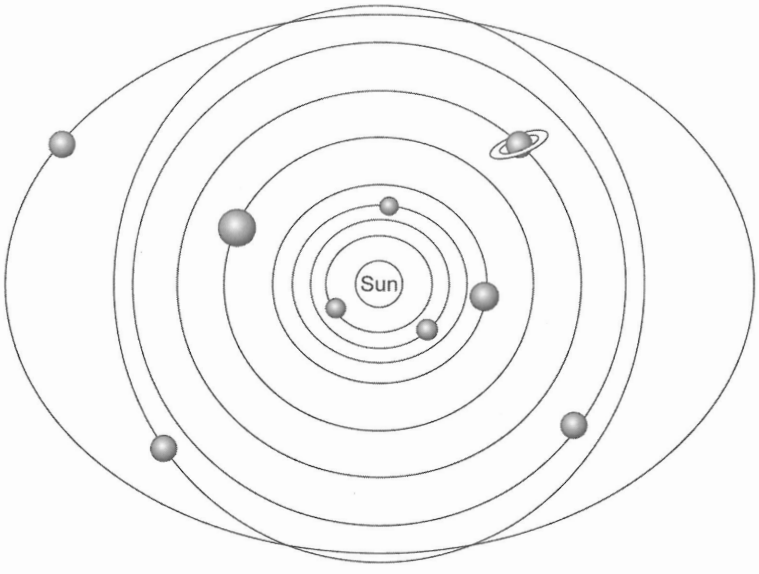
2 Read the information about Earth and our solar system at the top, then complete the diagram that shows Earth in relation to the Sun and the other planets. Note that the diagram cannot be drawn to scale. It is a schematic diagram because it is a model of reality.

a Number each planet and add a key.

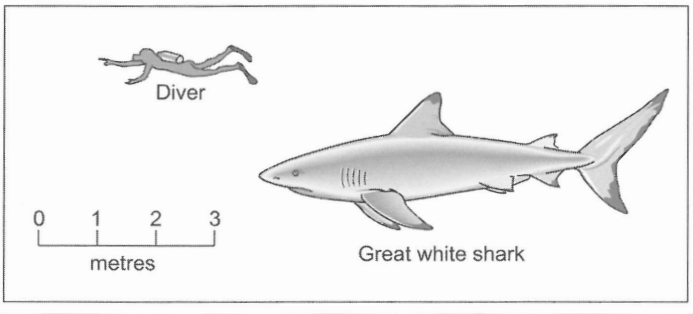
b Add a label giving the diameter of Earth.

c Add a title to the top of the diagram.

d Write along the bottom that the diagram is not to scale.



3 The small diagram below is a scale diagram. Explain what this is and why the schematic diagram of our solar system cannot be drawn to scale. (Hint: you should mention something about the distances and the sizes of the planets compared to the Sun. For example, Pluto's diameter is only 2300 kilometres.)



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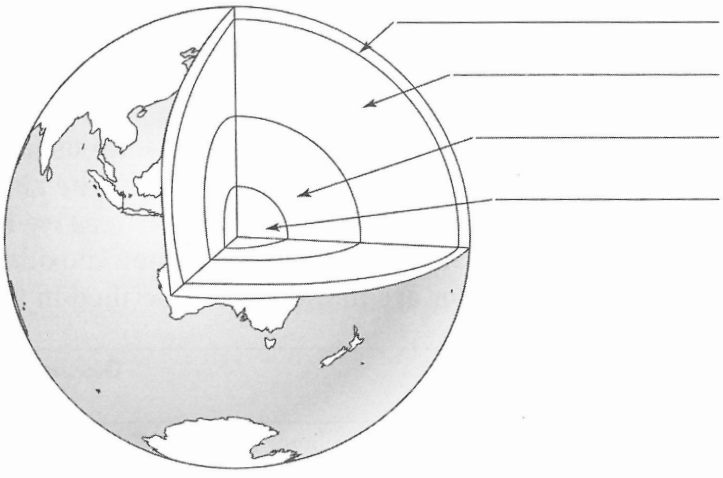
Earth's layers

Millions of years ago Earth was molten rock. Then the outer surface cooled and became solid. However, under the top layers, it still remains very, very hot. Scientists have been able to learn about the different layers that make up Earth's mass by studying earthquakes and volcanoes.

4 Use the following information to label Earth's different layers on the diagram next to the table. Add depths and temperatures.

Layer	Descripton
Crust	The hard and rocky outer surface of Earth, which is thinner under the oceans. The crust's thickness ranges between 6 and 75 kilometres. It is made up mostly of water, and different types of rocks and minerals. Temperatures on the crust's surface range from below zero to about 50°C, and up to 870°C near the boundary with the mantle.
Mantle	The rocky layer that extends just over 2800 kilometres. It is made up of silicon, oxygen, magnesium, iron, aluminium and calcium. The part of the mantle closest to the surface is rigid, but lower down it flows slowly – a few centimetres each year. Temperatures range for 1000°C at the boundary with the crust to 2200°C at the boundary with the core.
Outer core	The molten layer made up mostly of iron and nickel that surrounds the inner core. Its thickness averages around 2280 kilometres. Temperatures range between 2200 and 4500°C.
Inner core	The centre of Earth made up of solid iron and nickel. It is extremely hot with temperatures over 4500°C up to 7200°C. It is also under great pressure. Its radius is about 1230 kilometres.

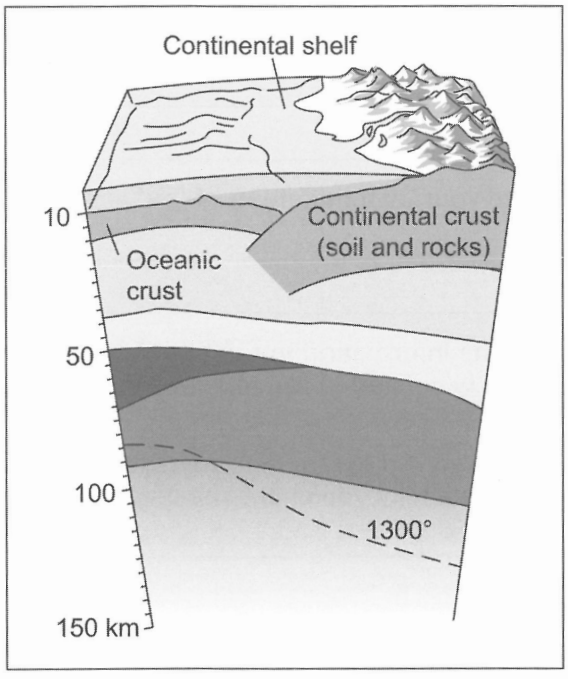


5 Choose an appropriate shade for each layer and colour the diagram. Also colour the features on the outer surface – the oceans and the continents. (Hint: think about the temperatures of the different layers when choosing your colours.)

Earth's crust

6 Examine the diagram of Earth's crust at right then fill in the gaps in the passage below.

The crust is the _____ layer of Earth. It consists of two parts. One part is the _____ _____, which is the land that we can see. It is made up of _____ and _____. The other part is the _____ crust. This is _____ than the continental crust, and lies under the world's _____ and seas. The continental crust contains landforms such as _____, hills, _____, rivers, valleys, flood _____ and coastal cliffs and _____.



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3 Above Earth's surface

Planet Earth can be divided into four main layers of different temperatures and material – the crust, mantle, outer core and, at the very centre, the inner core. Similarly, the air that surrounds Earth is made up of layers of gases that have different temperatures. This air is part of the atmosphere, which has five sub-spheres at different altitudes above Earth.

DEFINITIONS

gases are substances that are not either solids or liquids, such as steam and clouds – many are invisible, such as much of the air around us

the **atmosphere** is the blanket of gases that surrounds Earth to an altitude of over 1200 kilometres

altitude is the increase in height going away from Earth's surface at sea level

Earth's atmosphere

When we think about what surrounds us, we usually say 'the air'. Sometimes we describe the air as 'fresh', at other times it might be 'polluted'. The air, in fact, is part of Earth's atmosphere. Without the atmosphere there would be no life on Earth as we know it. The troposphere is 78% nitrogen and 21% oxygen, with tiny amounts of argon, carbon dioxide, water vapour and other gases. The five layers of the atmosphere in order are named and described in the table below.

Layer of the atmosphere	Description	Altitude (approximate)
troposphere	The bottom layer closest to Earth's surface where things live. It contains about 50% of the air we breathe. Also contains all the clouds we see and the weather we experience.	From sea level up to about 16 km
stratosphere	The next layer where the air is thinner because there are fewer gas particles than in the troposphere. Contains a further 40% of our air. There are no clouds here. Also contains the ozone layer.	From above 16 km to about 50 km
mesosphere	Gas particles become thinner and thinner in this layer. Shooting stars burn up here.	From above 50 km to about 80 km
thermosphere	Fewer gas particles and very little effect on Earth's air.	From above 80 km to about 600 km
exosphere	The very outer layer with only a sprinkling of gas particles, many of which escape to outer space.	Above 600 km to just over 1200 km

1 What is the suffix common to each layer of the atmosphere as well as the atmosphere itself? Why do you think this suffix is used?

2 Write your own definition of 'air'.

3 Use the information in the table and at the top of this page to complete the diagram of Earth's atmosphere that has been started for you over the page. Label each layer with its name, altitude and a very short description. Give your diagram a title.

4 Colour your diagram. For the layers of the atmosphere, choose no more than 2 colours, and use shades of these. State why you chose the colour/s.

5 Your diagram is not to scale. Explain why this is so.

6 What conditions in the troposphere make it vital to living things?

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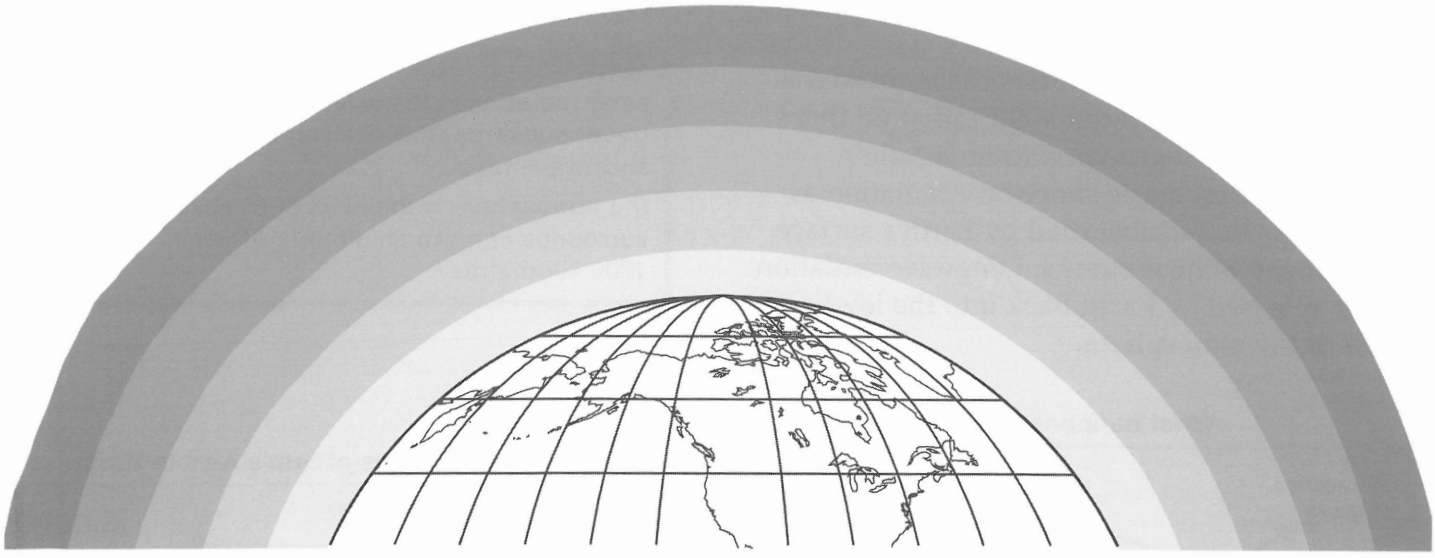

7 Does the diagram mean the atmosphere only covers the top of the northern hemisphere? Explain.

8 Did you use smooth lines between each layer?

a If YES, explain why this might be misleading. What would be a slightly better choice?

b If NO, explain why you chose to draw the borders between the layers the way you did.

The ozone layer

The stratosphere contains a small amount of ozone gas. This gas is made up of molecules each containing three atoms of oxygen – its chemical formula is O_3. (The formula of oxygen gas is O_2.) The ozone layer blocks most of the harmful ultraviolet (UV) radiation from the sun. There is evidence that the ozone layer is getting thinner due to chemical emissions from human activity on Earth.

9 In red on your diagram, add a narrow strip to the top of the stratosphere and label it the ozone layer.

True or false

10 State whether each of the following statements is true or false. For those that are false, rewrite them so they are true.

Statement about the atmosphere	T or F	Correct statement
The atmosphere surrounds Earth.		
The exosphere affects people the most.		
The mesosphere contains many clouds.		
The ozone layer is in the troposphere.		
The ozone layer blocks ultraviolet light.		
The ozone layer is getting thicker due to human activities. This is causing less sunlight to reach Earth.		

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4 Heating places on Earth

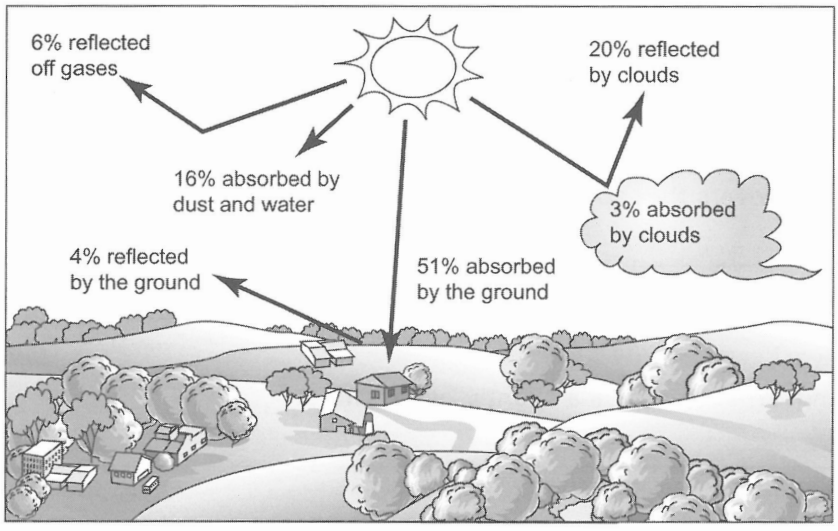
Solar radiation from the sun is the main source of Earth's energy. Without solar radiation there would be no plant growth and no weather systems. The sun emits shortwave radiation as light energy that is absorbed by Earth's surface and changed to heat energy. Longwave radiation is then released by Earth back into the lower layers of the atmosphere.

DEFINITIONS

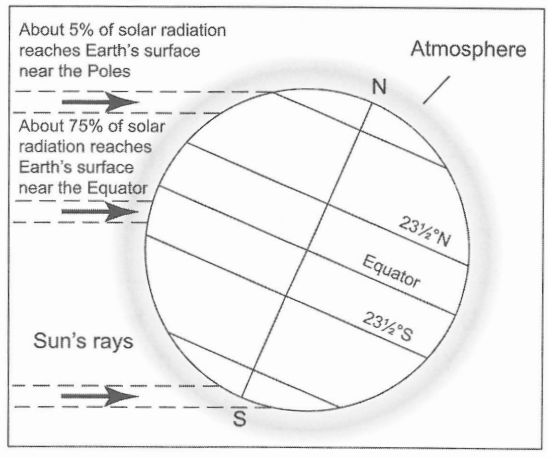
solar radiation is the waves of light energy that travel outwards from the sun in the form of shortwave rays

the **atmosphere** is the blanket of gases that surrounds Earth to an altitude of over 1200 kilometres

What happens to solar radiation?



Angles of Sun's rays to Earth



1 Use all the information above in both the text and the diagrams to fill in the gaps in this passage about how Earth is heated.

Heat energy from the _____ travels to Earth as short_____ _____. These rays do not directly _____ the atmosphere. Some rays (about 20%) are deflected by _____, which also _____ about 3%. About _____% of solar radiation is reflected by gases in the _____, and 16% is absorbed by _____ and _____. This leaves around _____% to directly hit Earth's surface. About 4% of this direct solar radiation is then _____ back into the lower layers of the _____. Earth is tilted at an angle of _____ degrees. This means that an area on Earth at low latitudes (closer to the _____) receives _____ heat than the same area at _____ latitudes (near the Poles). The amount of solar radiation received is the _____ all over Earth. However, the rays hit Earth's surface at _____ angles in different locations. They hit at about 90 degrees at the _____. They therefore have _____ atmosphere to go through so _____ energy directly hits the ground. Near the _____ the sun's rays strike at a much lower _____. They travel through a longer section of atmosphere, which scatters the heat energy, making these places _____ than those _____ the Equator.

2 Explain how the tilt of Earth on its axis affects the amount of solar radiation received at different times of the year where you live. On a separate sheet draw a diagram to illustrate your explanation. (Hint: think about the four different seasons of the year.)

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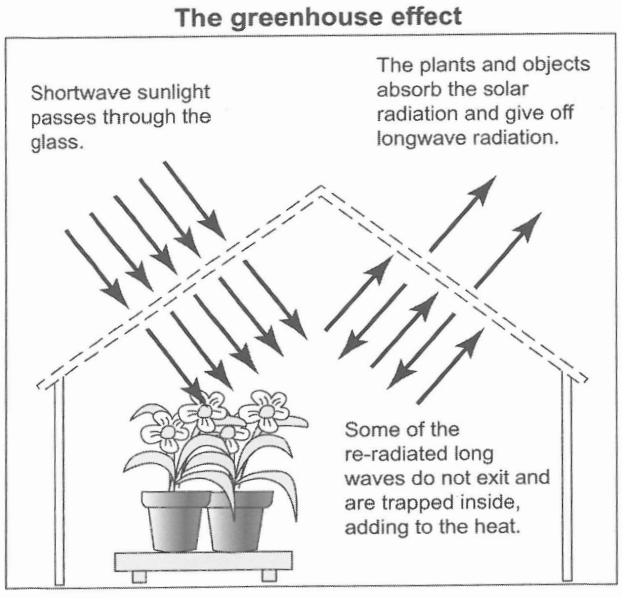
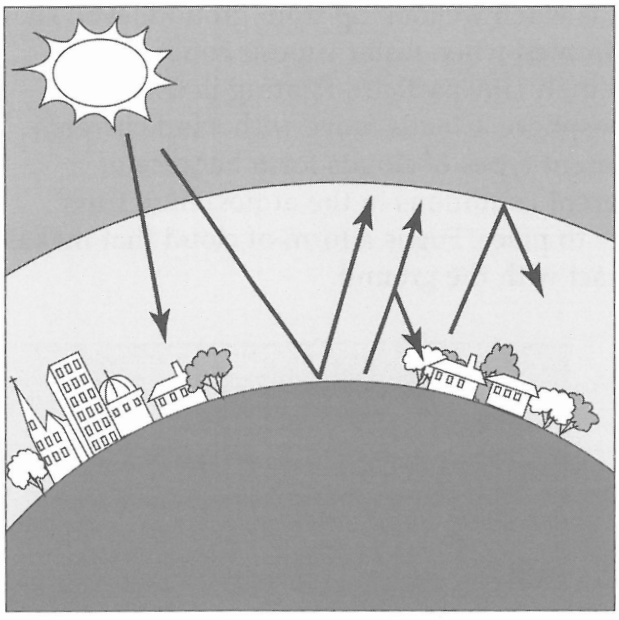
The greenhouse effect

Some of the heat that is reflected from Earth's surface is trapped by gases in the lower layers of the atmosphere. These gases are known as 'greenhouse' gases. The main greenhouses gases are water vapour, carbon dioxide, methane, nitrous oxide and ozone. Certain human activities add to the levels of most of these naturally occurring gases.





3 Describe how the left diagram above illustrates what happens to heat in a greenhouse. What does this mean for plants growing inside compared to outside?

4 A similar 'greenhouse effect' occurs on Earth as some of the longwave radiation is reflected back from the ground and then trapped in greenhouse gases in the lower atmosphere. Label the diagram at top right to show this effect.

5 Many scientists now believe that the greenhouse effect is being 'enhanced'. What does this mean? (Hint: enhanced = increased.)

6 State how each of these situations could enhance the greenhouse effect:

a increased use of cars, power stations and factories:

b increased population needing more agricultural products:

c clearing of forests, especially by fire, for human development:

7 If the enhanced greenhouse effect continues, what do many scientists and geographers believe is the likely result for the globe over the next 100 years or so? What is the name given to this?

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5 Cloudy skies above

Pictures from satellites circling above show Planet Earth as a beautiful blue ball that is surrounded by trails of clouds. We can also see clouds when we look up from ground level. They are formed when water vapour condenses on extremely tiny particles floating in the troposphere. Clouds move with wind currents. Different types of clouds form because of different conditions in the atmosphere from place to place. Fog is a form of cloud that makes contact with the ground.

DEFINITIONS

water vapour is the gas that forms when water has been evaporated from Earth's surface

to **evaporate** is to change from liquid to gas through heating

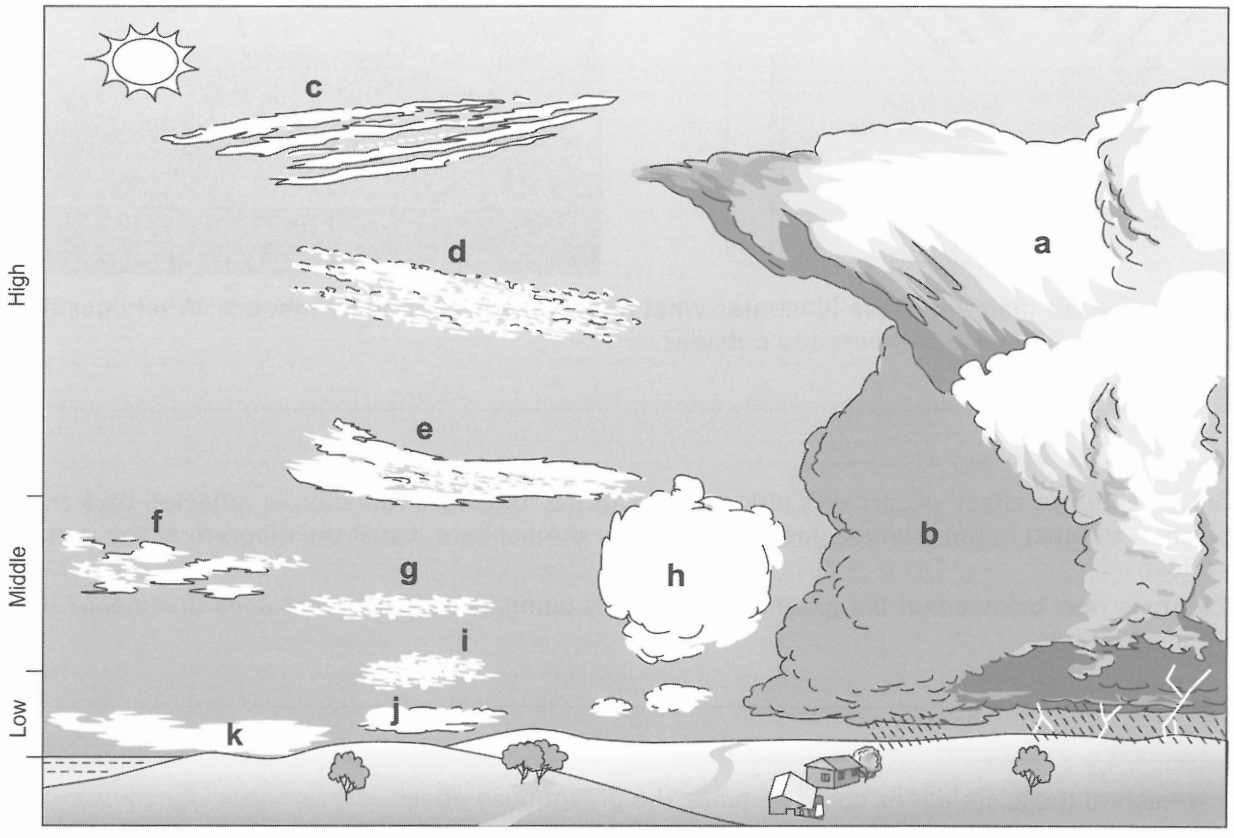
to **condense** is to change from gas to liquid or solid when warm air meets cold air

the **troposphere** is the layer of the atmosphere closest to Earth's surface

altitude is height above sea level

meteorologists are scientists who study and predict the weather

insulation is a protective cover



1 Read the section over on 'Classification of clouds' including the table, which is organised alphabetically. Name the clouds **a–k** in the diagram above. Add a scale to the vertical axis. Give your diagram a title.

2 Your completed diagram is a schematic diagram. What does this mean? Explain how it helps you understand more about clouds.

3 Clouds indicate weather conditions, so meteorologists have grouped them into four main types. Complete the table to match the type with cloud names. (Hint: there are 3 in the first category and only 1 in the last. Also do not include 'fog' as an example.)

High clouds (bases > 6 km)	
Middle clouds (bases 2 to 6 km)	
Low clouds (bases < 2 km)	
Extends through all altitudes	

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Classification of clouds

4 Provide examples for the table below after reading the information in this section.

Clouds are classified into a system that uses Latin words to describe the way they look as seen from the ground. There are four main source terms:

Latin root	Translation	Example
cumulus	heap or pile	
stratus	layer	
cirrus	curl of hair	
nimbus	rain	

A prefix based on altitude is then added to seven cloud types. There is no prefix for low clouds.

- Names containing the prefix 'cirr-' (as in cirrus clouds) are located at high levels.
- Names with the prefix 'alto-' (as in altostratus) are found at middle levels.

Cloud type	Appearance	Altitude range
Altocumulus	Medium-sized, puffy patches, often in wave-like bands	From 2000 to 6000 metres
Altostratus	Thin, continuous sheets	From 2000 to 6000 metres
Cirrocumulus	Small puffy patches	Above 6000 metres
Cirrostratus	Thin and feathery, usually above a thundery cloud	Above 6000 metres
Cirrus	Thin, wispy strands	Above 6000 metres
Cumulonimbus	Huge and towering, often with a flat, dark top associated with electrical storm activity.	From 2000 to over 20 000 metres
Cumulus	Puffy and piled up on a flat base	Below 2000 metres
Fog	Very low mist, especially on valley floors	Making contact with the ground
Nimbostratus	Dark and flat-based, producing rain	Below 2000 metes
Stratus	Even, flat base, thick to thin with ragged edges	Below 2000 metres
Stratocumulus	Broad and flat base with a puffy top	Below 2000 metres

Clouds affect temperature

5 In the space below, draw and label 2 diagrams of the same size side by side. Give each a title. (Some words are provided in brackets to help you with the labels.)

a In the left diagram show how cloudless skies can cause high day temperatures but very low night temperatures. (solar radiation, escape)

b In the right diagram show how a blanket of clouds can make the night warmer than when the sky is clear. (blanket, heat, insulation)

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6 Rain formation

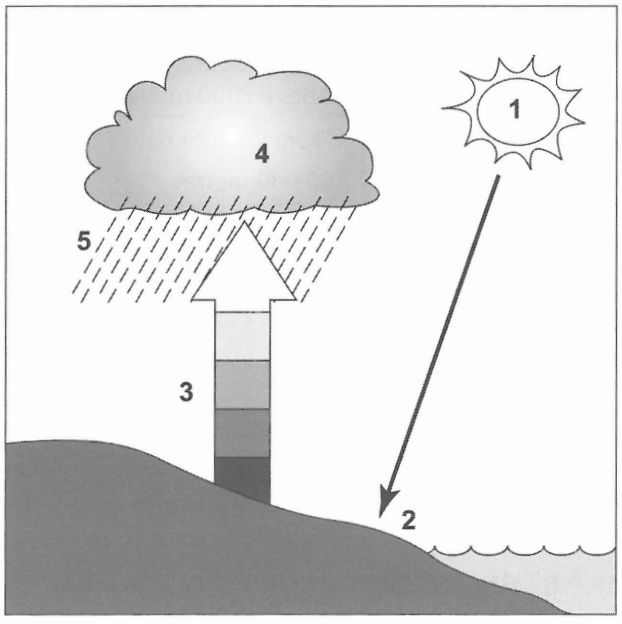
Life on Earth depends on water for survival. Water reaches Earth as precipitation. Rain is the main type of precipitation and falls nearly everywhere over the globe except where it is very cold and turns to a solid. There are three main ways that rain forms. These depend on the local conditions of temperature and wind, but all result from air rising. Two ways are shown on this page, and the third is shown over the page.

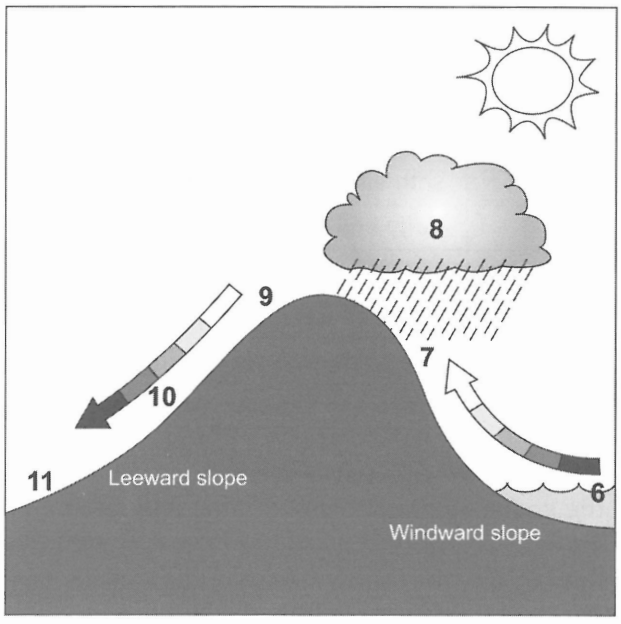
Facts about air and water

- Warm air rises.
- Warm air picks up water.
- Warm air can hold water.
- As air rises it cools.
- Air that is getting cooler releases water.
- Water falls from cooling air.
- Heat is released when it rains.

DEFINITIONS

precipitation is all types of water that reaches Earth either as a liquid (rain, dew) or solid (snow, hail, frost)
local conditions are felt in a fairly small area of the globe up to a few hundred square kilometres
convection occurs when the Sun heats the ground causing air to rise rapidly
orographic means being formed by rising over a high landform
a **cold front** forms where a cold air mass collides with warmer air and tries to replace it





1 The diagrams above show two different ways that rain forms. Use the scrambled steps below and match them with the correct number – the left table corresponds with the left diagram, and the right table with the right diagram.

	Large clouds develop directly above the rising air
	The Sun heats up the ground rapidly
	Heavy rain and thunderstorms result due to rapid cooling where water droplets collide with each other and fall
	The ground heats up through convection and the air rises rapidly taking with it evaporated water
	The quickly rising air cools rapidly

	Dry air slides down the slope and starts warming up again
	Evaporation occurs over the sea
	Water vapour in the air condenses and cools forming rain clouds
	Cold air becomes dry
	Moist air is forced to rise over a hill or mountain, cooling at a rate of 0.6°C per 100 m
	Clouds disappear and the air is dry with warm winds, creating a rainshadow

2 One diagram shows Orographic Rain and the other shows Convectional Rain. Use the definitions at the top of the page to give each diagram its correct title.

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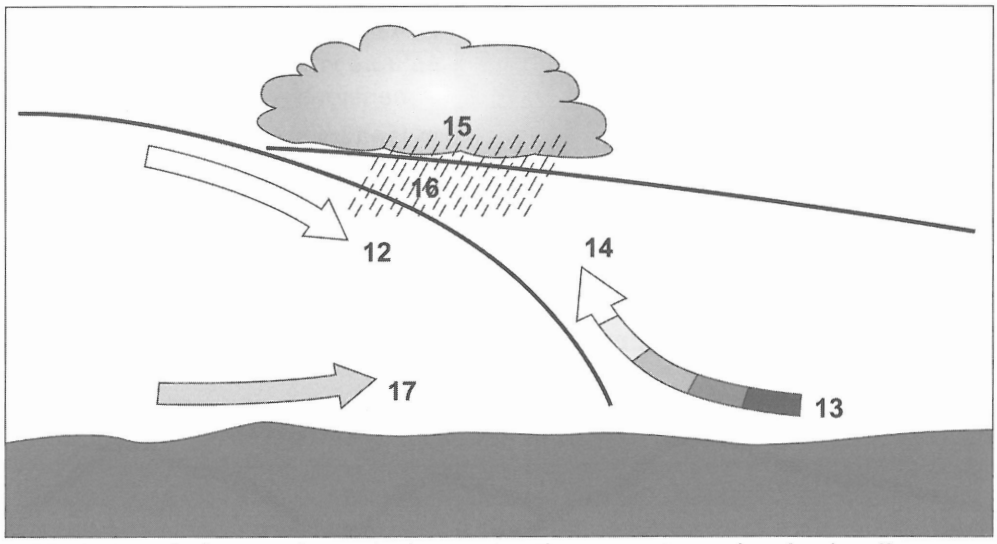
Frontal rain

When large bodies of air of different temperatures meet they form a front. They do not merge (combine) because one air mass is warm and one is much colder. Frontal rain is also known as cyclonic rain. Frontal rain can be caused by a cold front or a warm front.



3 Use the scrambled steps below and match them with the correct number in the diagram above. Add a title to the diagram.

	The air is forced to rise and it cools
	The cold air is heavy and pushes forward along the ground
	Further upward movement in clouds causes rain
	Frontal movement
	Water vapour condenses to form clouds
	The warmer air is lighter so is forced to rise above the cold air mass

4 Rain can also result from a warm front. On a separate sheet draw and label a diagram that explains this type of frontal rain.

True or false

5 State whether each of the following statements is true or false. For those that are false, rewrite them so they are true.

Statement about rain	T or F	Correct statement
Rain is one type of precipitation.		
Rain is not formed when air rises quickly.		
A rainshadow forms on the windward slope of a mountain.		
Cold air is drier and lighter than warm air.		
Convectional rain occurs more often in local conditions that are cool and dry.		
Thunderstorms and heavy rainfall are often the result of convectional rain.		

6 What is a schematic diagram? Explain how each of the three diagrams in this topic is a 'scheme'.

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7 A RIVER SYSTEM

If you were flying high in outer space looking down on Earth, you could see a pattern of veins branching out over the land surface. This pattern is formed by the thousands of different streams and rivers that make their way from highlands to larger bodies of water – inland lakes, and the seas and oceans. As soon as rain falls on the ground or snow melts, water starts moving downhill. At the source the flow is only a trickle, then it becomes bigger and stronger as it travels down steep slopes to flatter land. The river is joined by tributaries, which add to its volume.

DEFINITIONS

the **source** is where something starts

a **tributary** is a stream joining and adding its flow to a larger stream or other body of water

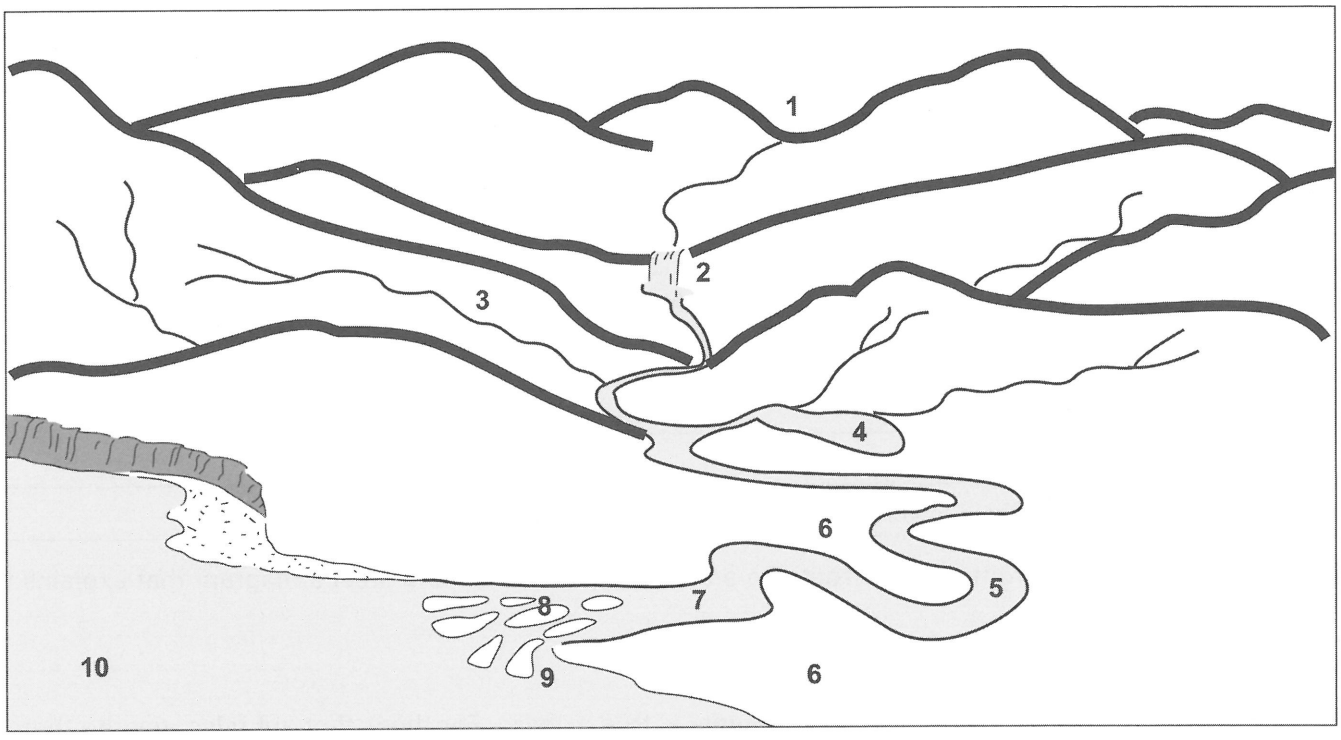
volume is the total amount of something in three dimensions – length, width and depth

meanders are winding parts of rivers on flat land

an **estuary** is the part of a river close to the sea where fresh and saltwater mix

the **mouth** of a river is where it spills into the sea

a **delta** is a nearly flat plain deposited between branches of the mouth of a river



1 Match the letter on the diagram above with the correct term.

Number	Part of a river system
	delta
	estuary
	flood plain
	lake
	meander
	mouth
	sea
	source
	tributary
	waterfall

2 Colour the main river blue from its source to its mouth.

3 'Delta' is actually a letter of the Greek alphabet that is shaped like a triangle (Δ). Why do you think the term is used for the land at the mouth of a river?

_____

_____

_____

4 Shade in red the area where you would locate a town. Why did you choose this?

_____

_____

_____

_____

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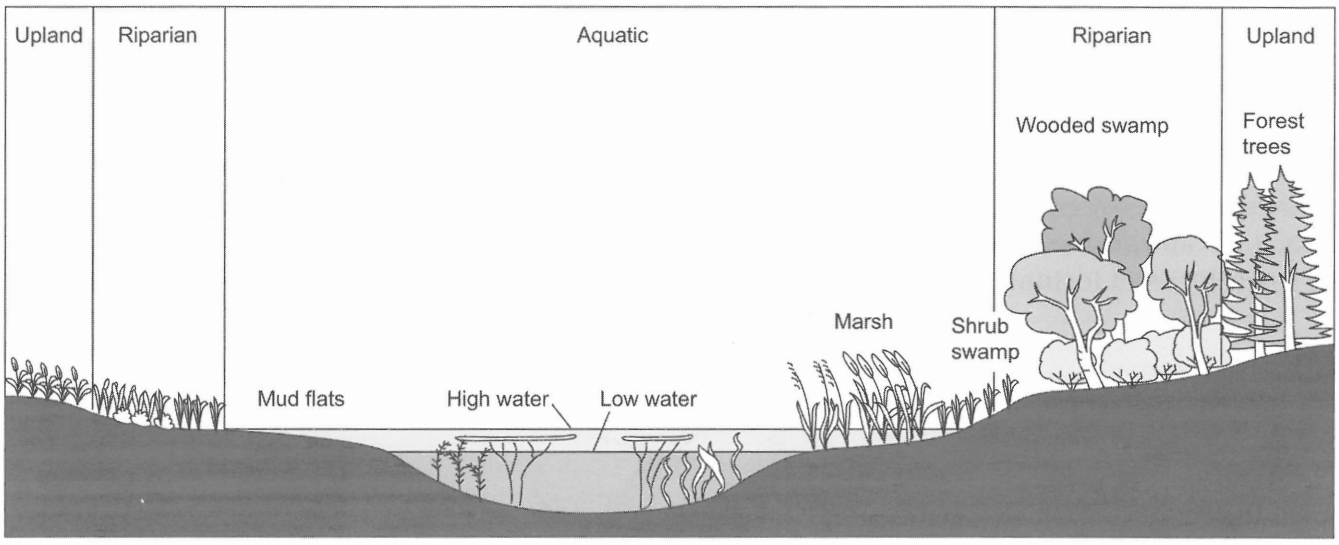
Wetlands

There are many points in a river system where the water and land meet. These areas are called 'wetlands'. Some wetlands hold water all year round, while others dry out for short or longer periods. Wetlands have plants and soils that are suited to the change between wet and dry. There are three main zones in a wetland.



5 Complete this passage about the environment and importance of wetlands by using all of the words provided. Note that you may need to use a word more than once.

absorb, breeding, drier, fish, forest, higher, insects, land, limit, natural, riparian, river, shallow, sponges, upland, vegetation, water, wet, wetlands, wettest, year, zones

Wetlands are _____ areas where the _____ meets the land.

Most wetlands have three distinct _____. The aquatic zone is the _____ zone. It can be quite deep and contain water all _____ round, or it could be very _____ with no open water. Many different _____ plants grow in this zone. The second part of the _____ is the _____ zone. This is a strip of _____ on the bank of a _____ where plants usually grow well under _____ conditions. The _____ is the last of the wetland zones. This is the _____ land bordering the wetland where _____ trees and grasses grow.

_____ are important because they act like giant _____ that soak up water during _____ periods. The wetland _____ slows down running _____ and can help _____ floods. The upland and _____ zones also help filter water as it moves from _____ to lower ground. The plants and soils in these two zones _____ chemicals and particles carried by the water.

Wetlands provide safe _____ grounds for many birds, _____ and _____ and they depend on this ecosystem for their survival.

6 Estuaries are a particular type of wetland. Research the life in an estuary, including plants, animals, fish and insects, and what influence the tides have on an estuarine wetland. On a separate sheet, create your own labelled diagram of an estuary. Is your diagram schematic like the river system one or a cross-sectional profile like the one above?

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8 THE MOVING EARTH

Most school atlases have a world map showing the main tectonic plates that make up Earth's crust. Scientists think the plates were caused by cracking when Planet Earth cooled millions and millions of years ago. The continents and oceans are floating on these huge plates. They rest on Earth's hot mantle. So far 7 large and about 20 smaller plates have been marked out and named. The plates are separated by fault lines. The movement between plates and along faults is jerky and can cause earthquakes. There are three main types of tectonic movement.

DEFINITIONS

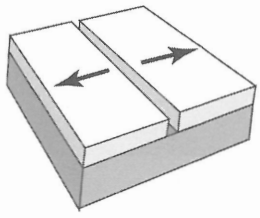
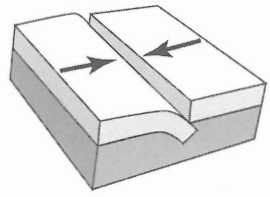
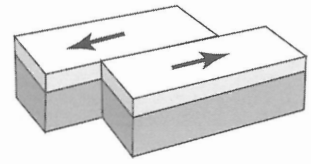
a **continent** is one of the seven huge land masses that Earth can be divided into

tectonic means something do with Earth's crust

a **tectonic plate** is a huge sheet of rock between 6 and 75 kilometres thick that lies under part of a continent or one of the oceans

a **fault** is a break in a mass of rock causing disturbance along the crack

subduction occurs when an oceanic plate slides under a lighter continental plate

		
Divergent boundary	**Convergent boundary**	**Transform boundary**
You can simulate divergent plate motion by placing your palms down with your index fingers touching. Now pull your hands apart.	You can simulate convergent plate motion by placing your hands together in front of you with your palms down. Gently slide your left hand downward and your right hand upward but keep them in contact.	You can simulate transformational plate motion by placing your hands together in front of you with your palms down. Now slide your right hand away from you and your left hand toward you in the same plane.

1 In the empty spaces in the table above describe the three different types of tectonic plate movement.

2 The diagrams are block diagrams. What does this mean? How do they help your understanding?

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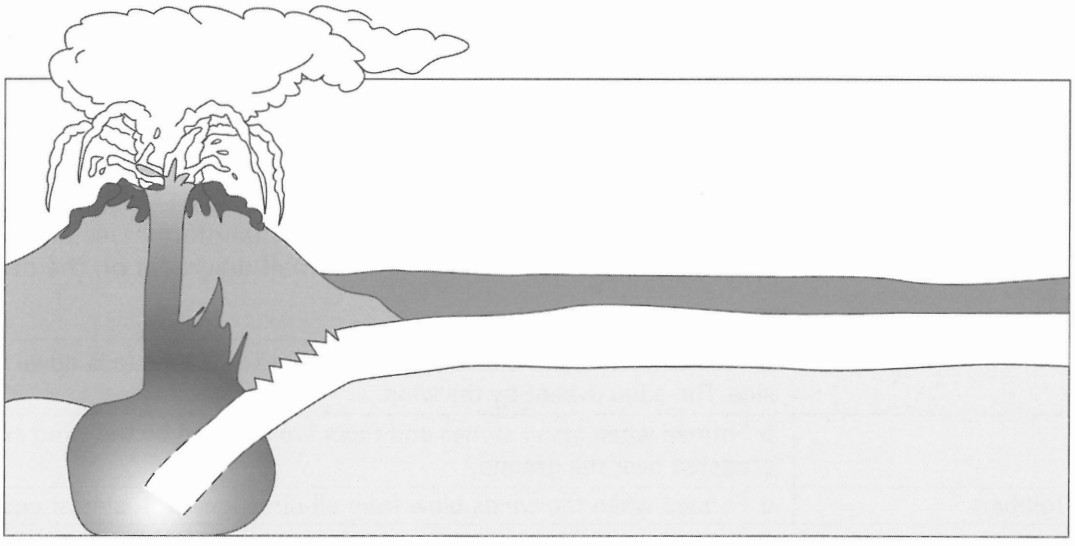
3 Explain how earthquakes can result from any of the types of tectonic movement. (Hint: use the word 'fault' in your answer.)

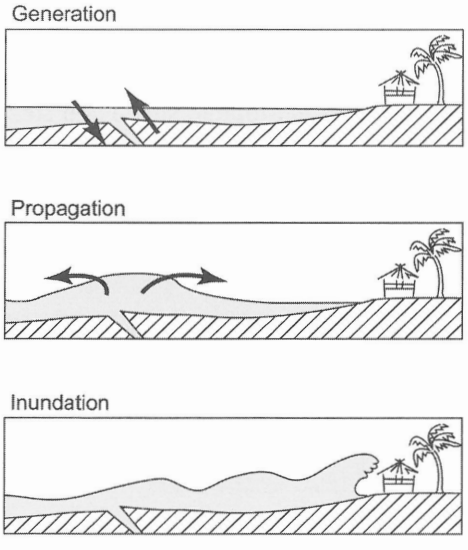
4 Another tectonic event that may occur near the location of an earthquake is a _____.

5 Add the correct labels to the diagram below. Give it a title.

ash cloud, continental plate, crater, Earth's crust, lava flow, magma chamber, oceanic plate, subduction zone, volcano



6 The diagram below shows a natural hazard that can result from an earthquake. Add a title with the name of the natural hazard. Write a description of how it is caused.



7 **Research:** Which tectonic plates caused the devastating tsunami of 26 December 2004? Describe what happened.

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9 DESERT LANDFORMS

Across the globe there are many dry areas that receive 250 millimetres or less total annual precipitation. Deserts can be hot or cold. In hot deserts the wind and rain form the land into different shapes. Although there is not much water, when it falls it is usually in heavy storms. Rain strikes the hard ground. It does not have time to soak in so runs off in wild torrents causing a lot of erosion. The wind in deserts is usually strong and contains millions of sand or rock particles, which also cause erosion. Eroded material is then deposited. Desert landforms are the result of this erosion and deposition.

DEFINITIONS

a **torrent** is a large volume of fast-moving water

precipitation is all types of water that reach Earth's surface either as a liquid (rain, dew) or solid (snow, hail, frost)

erosion occurs when the land is worn away by running water and blowing wind

deposition occurs when eroded material is carried away and then settles in a new place

1 In the list below join the correct description with the name of the desert landform. Use a different colour or type of straight line for each join. (Hint: you might like to refer to the small diagrams on the next page for a visual guide.)

Alluvial fan	**a** Formed when sand blasts up the windward side then falls down the leeward side. The edge is bent by the wind.
Butte	**b** Formed when small stones and rocks are dropped by the wind and left scattered over the ground.
Stony plain (gibber)	**c** Formed when the winds blow from all directions with almost equal force.
Arch and mushroom rock	**d** Formed when rainwater collects in holes on a very flat and low surface.
Longitudinal dune	**e** Formed when very hilly country is heavily eroded by severe storms and flash floods. Provides good hiding places.
Playa lake	**f** A sand dune that forms as a ridge parallel to the wind direction.
Star dune	**g** Formed when quick-flowing streams travel through narrow gorges and pick up loads of sand and rock pieces. When they start to reach flatter ground the streams spray the eroded material out in a fan shape.
Badlands	**h** Formed when a plateau erodes to leave a smaller flat-topped shape and steep slopes.
Mesa	**i** Similar to a mesa but with a smaller flat-topped area.
Barchan dune	**j** A sand dune that forms as a ridge at right angles to the wind direction.
Transverse dune	**k** Formed by sand blasting where the wind throws material against rocks that gradually wear away into interesting shapes.
Wadi	**l** A dry river bed that has been eroded by flash floods.

2 Some desert landforms are caused by 'flash floods'. Explain what these are and how they affect the desert landscape.

3 **Puzzle**: Pancho Villa (1878–1923) was a Mexican who used the desert badlands of southern USA. What was Pancho and why did he use the badlands? Why do you think this landform was given the name 'badlands'?

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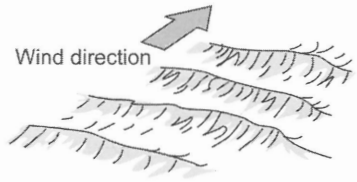
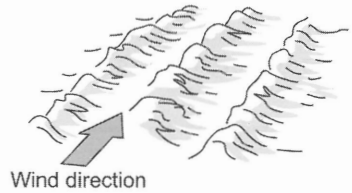
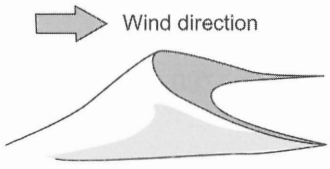
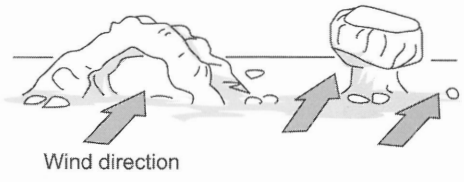
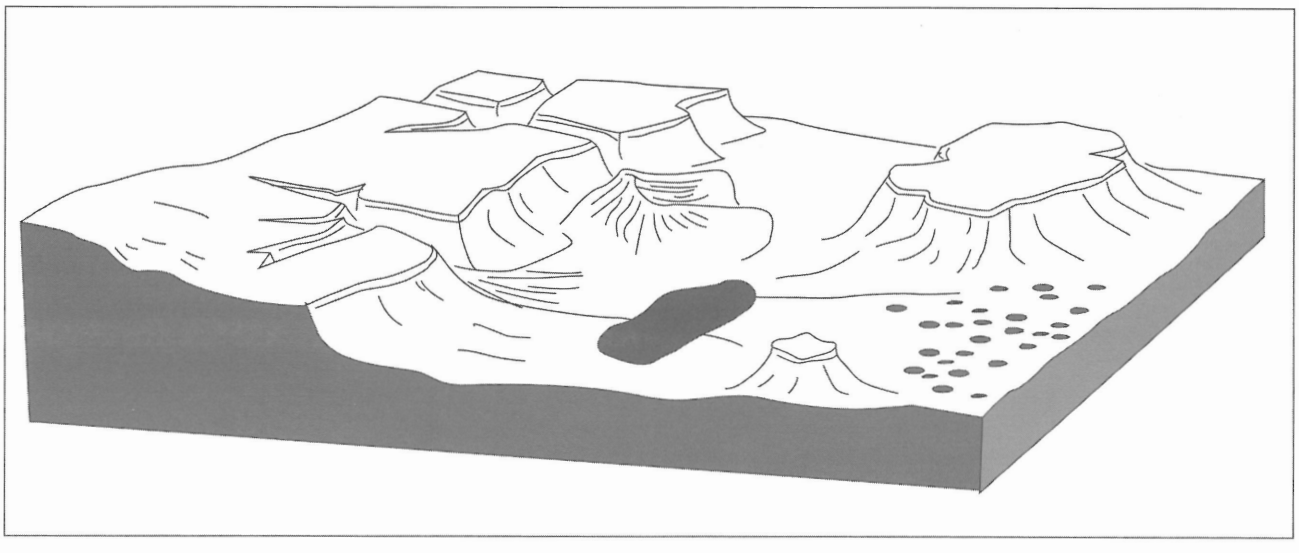
4 Use the answers you worked out for Question 1 to complete the name and description of each desert landform in the table and also on the diagram below. Add a title and colour to the bottom diagram. (Hint: the landforms in the table are not repeated in the diagram.)

Desert landform	Name	Description
		
		
		
		



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10 From natural to built

A block diagram is a 3D model of an area.

In Geography, block diagrams are especially useful in showing features of:

- different habitats and environments
- volcanoes, canyons, river valleys, glaciers and earthquake faults.

The sequence of three block diagrams in this topic represents the change over time that has occurred in many places in Australia. As the population has grown, the built environment has replaced more and more of the natural and managed environments.

DEFINITIONS

3D means a diagram is three-dimensional – length + width + height (or depth)

the **built environment** is what people have made and constructed

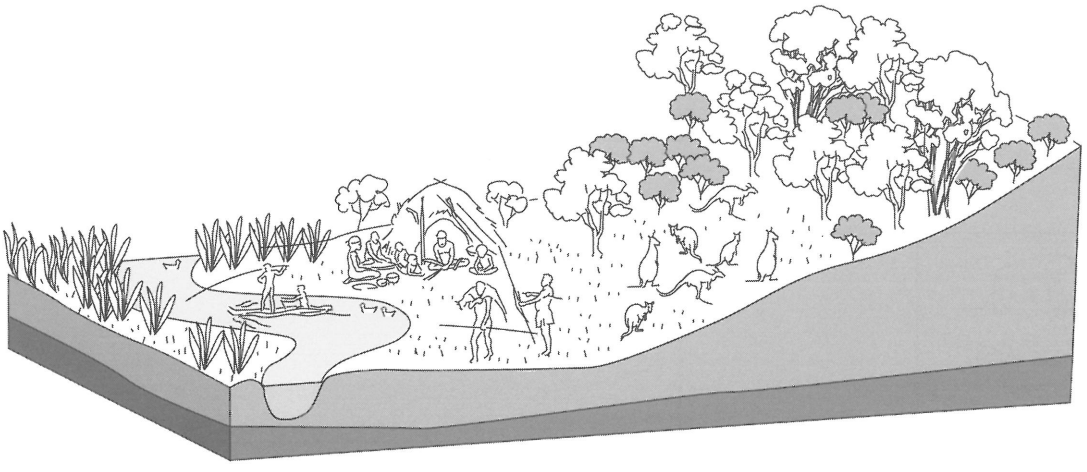
the **natural environment** is all the living and non-living parts of Earth that have been formed by nature (also called the biophysical environment)

a **managed environment** is one where humans have changed some aspects of the natural environment without dominating it with built structures, for example national parks and some farming land

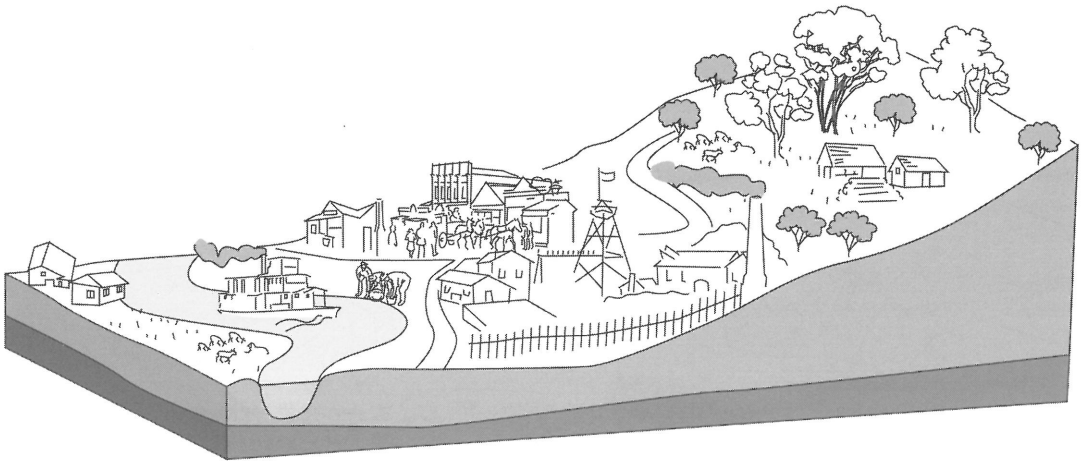
1 Why is a block diagram similar to cutting a slice of layer cake? _____

_____

2 List the features of the first block diagram, which represents an area of Australia in 1780. At the end of the list, state what type of environment dominates.



3 Do the same for the second block diagram representing 1850.



4 What changes have taken place over the 70-year period?

_____

_____

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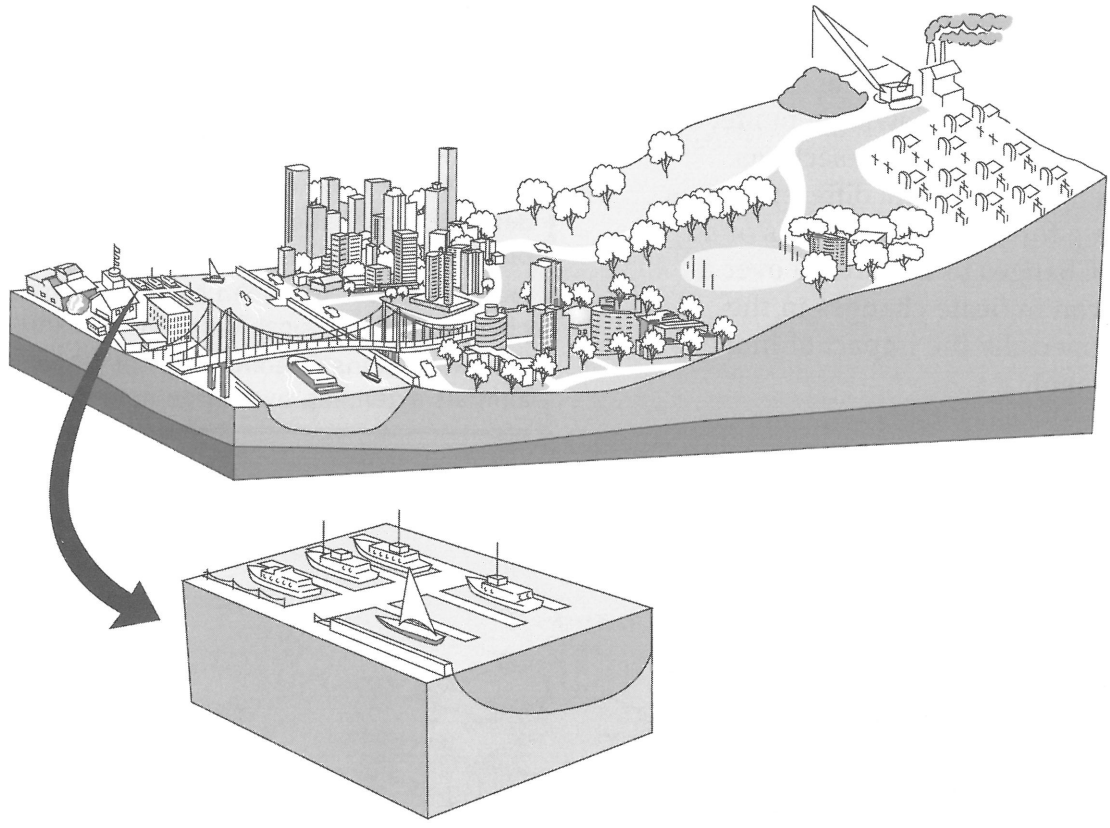

5 List the things in the block diagram for 2000 that are not in the previous two.

_____

_____

a What environment now dominates? _____

b Why has this changed occurred? _____

_____

6 Block diagrams show what sort of view? _____

a Is the view to scale? Explain.

_____

b What special feature has been added to the 2000 block diagram and how does this add to your understanding of the environment of the area?

_____

_____

7 Give two other ways that the change over time sequence for 1780, 1850 and 2000 could be shown and state how they would be different to the representation shown in the block diagrams.

1 _____

2 _____

8 Imagine what future changes will occur by 2050 and make a list of these. On a separate sheet draw a block diagram depicting the same area in 2050 that shows the changes you have listed. (Hint: trace the diagram above to use as a template. Erase the features you want to remove or change, then add your new features.)

Changes for 2050: _____

_____

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11 PROFILING VEGETATION AND SOILS

Profiles are diagrams that show the hidden parts of a geographic feature by viewing a model from side on. They are similar to cross-sections. Profiles are often used to show different layers of vegetation in a forest or the horizons of a particular soil. Changes to the profile over a period of time can indicate changes in the environment, especially the impact of humans on the environment.

DEFINITIONS

a **cross-section** is a 2D side-on view of the shape of the ground surface

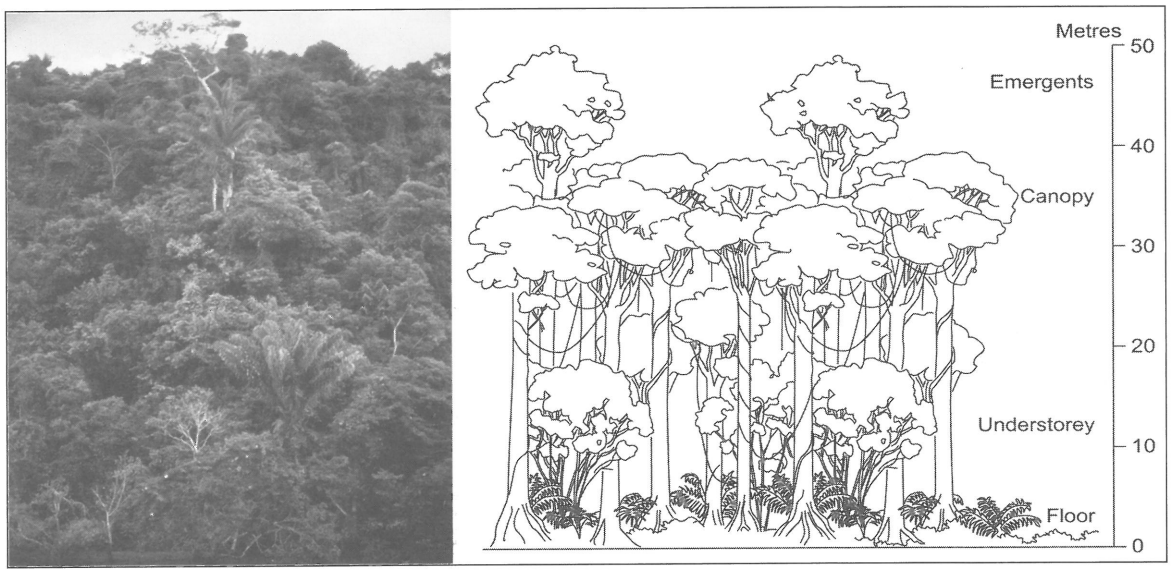
horizons are the layers of a specific soil

deforestation is the removal of forest trees and other vegetation

weathered rock has been broken down due to the physical and chemical forces of heat and water

a **biome** is a major community of plants and animals extending over large natural areas

1 Answer questions **a–c** about this combination photo and diagram.



a Why is it a profile and what it is profiling?

_____

_____

b Why have measurements been added?

_____

c In the right column, give a definition for each layer. Use the scale to help.

_____

_____

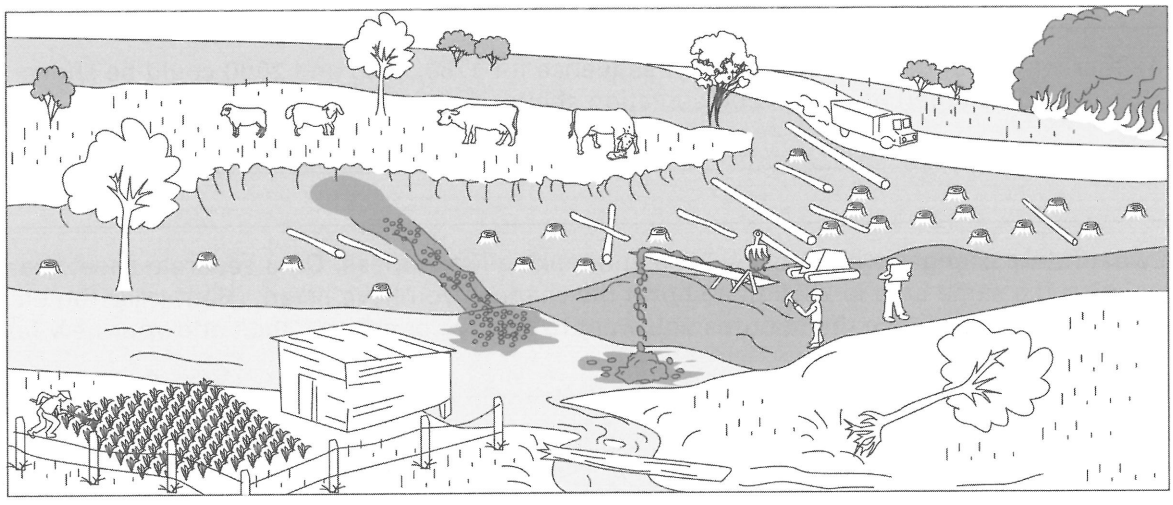
_____

_____

_____

_____

2 Human activities are gradually changing the rainforest profile. Add annotated labels to the diagram below, which shows some of the causes of deforestation and its effects.



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Tropical rainforest soils

Soil is the very top thin layer of Earth's crust, ranging from a few centimetres to several metres thick. Most of Earth's land surface is covered in soil of some type. Soil consists of:

- weathered rock pieces
- minerals
- organic matter
- gases
- water
- living organisms.

Soils in the world's different biomes have different horizons. They can vary in depth and also in the type of minerals, and amount of organic matter and water. However, the basic structure of all soil profiles is similar to the profile for a tropical rainforest below. The horizons are labelled O (organic matter), A, B and C.

3 Add these labels to the diagram at right.

Thin humus layer due to rapid decomposition
Solid parent bedrock that weathers quickly due to leaching
Thick leaf litter (dead leaves) that rots quickly
Clay particles rich in iron and aluminium
Lush vegetation with relatively short root system
Red layer due to mineral concentration of iron and aluminium
High rainfall is absorbed by plants or rapidly drains through the soil

4 Define these terms related to tropical rainforest soils:

Weathering is _____

_____

Leaching is _____

_____

Decomposition is _____

_____

Organic matter is _____

_____

Humus is _____

5 There are many tall trees in a tropical rainforest but their root systems are shallow compared to trees in cooler forests. What does the diagram tell you about why this is so?

_____

_____

_____

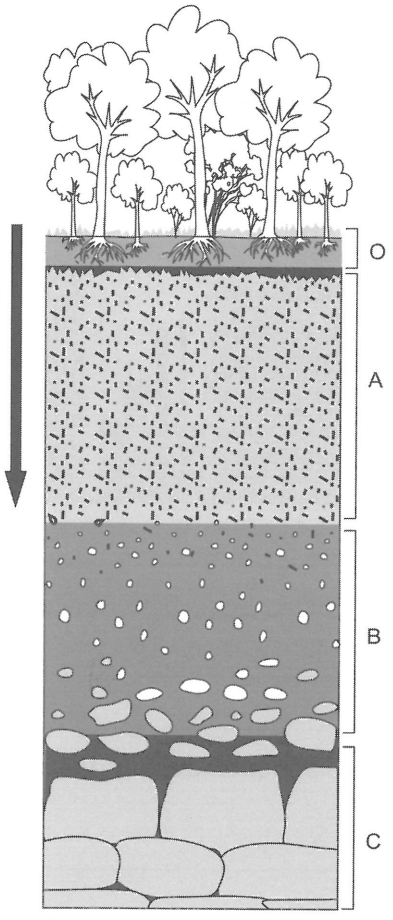
6 Explain why tropical rainforest soils are not as fertile as many other soils in the world even though there is a thick leaf litter horizon that decays rapidly.

_____

_____

_____

Profile of typical tropical rainforest soil



7 What is likely to happen to the soil if humans cause deforestation? What could be the result of this?

_____

_____

_____

_____

8 How have these 3 diagrams helped your understanding of the tropical rainforest biome?

_____

_____

_____

_____

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12 TRANSECTS SHOW CHANGE

Transect diagrams are similar to profiles in that they are cross-sections. They are made by geographers who observe the pattern of things on the ground or from details on a topographic map. A single-category transect shows the change that occurs in one main feature, such as the type of human activity across a city in an urban transect. A more detailed transect is divided into multiple categories organised into a labelled grid underneath the total length.

DEFINITIONS

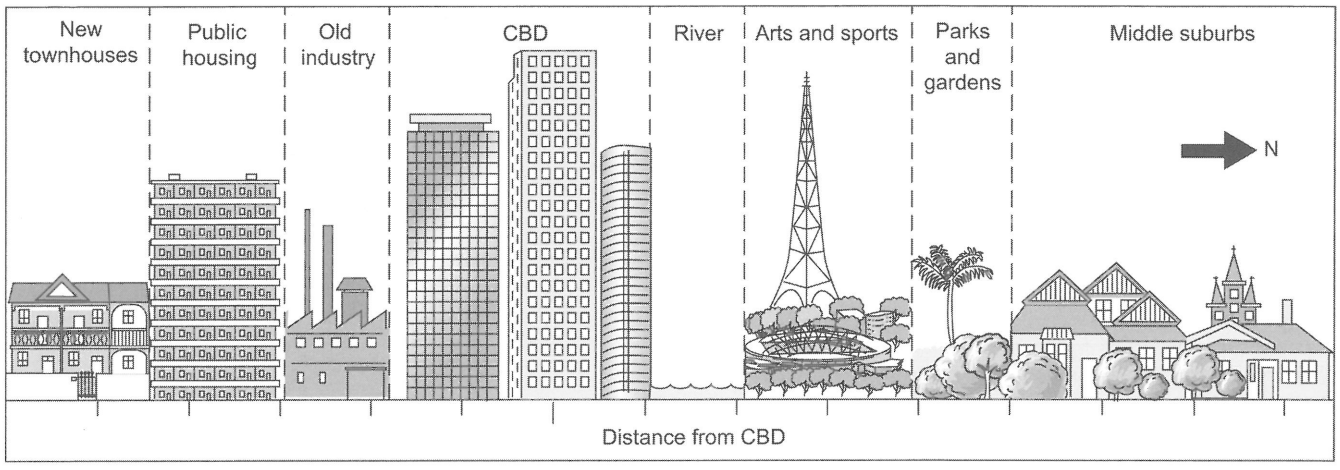
a **transect** is a straight line drawn between two geographic points that shows changes

a **cross-section** is a 2D side-on view of the shape of the ground surface

a **topographic map** is a detailed map of a fairly small part of a country with contour (height) lines as well as other human and natural features

the **CBD** is the central business district of a city

1 Examine the urban transect below and then answer the questions that follow.



a Why is this called an 'urban' transect?

_____

_____

b How many distinct landuse zones are there? Name them.

_____

_____

_____

c The old industrial zone is 2 kilometres from the highest building in the CBD. Complete the distance scale on the diagram.

d How many kilometres from the CBD are the new townhouses? _____

e What is likely to happen to the old industrial area? Why?

_____

_____

_____

2 The diagram at right is a transect too.

a Name the categories that change.

_____

_____

b What type of vegetation is at point X?

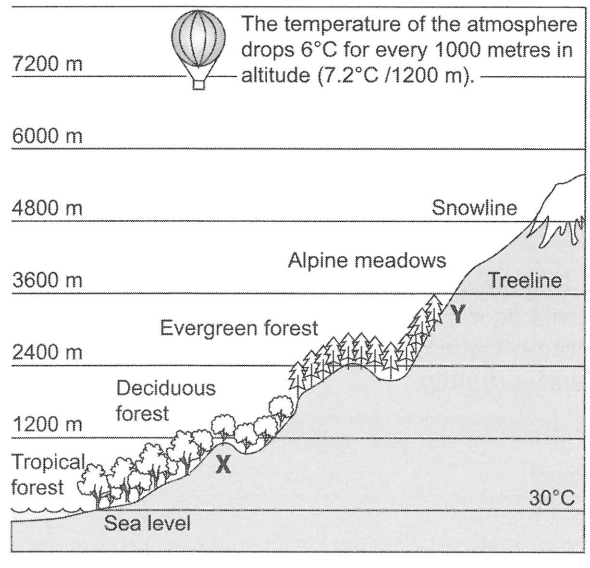
_____

c Complete the temperature scale. What is the temperature at point Y? _____

d Does this diagram depict a real place? Explain.

_____

_____



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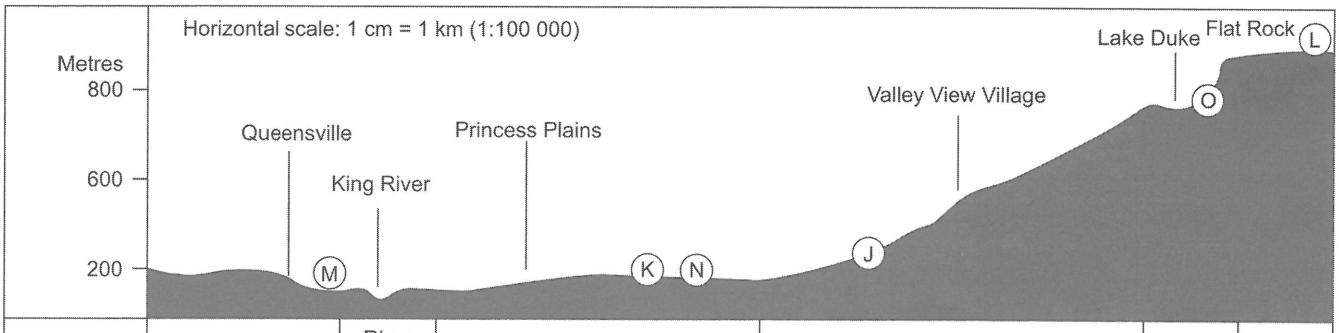
3 The previous diagram is a generalised model. What does this mean?

4 Write a statement that summarises the effect of altitude on vegetation.

5 How is the transect diagram below more detailed than the previous two?



Landform	Undulating plain	River flood plain	Undulating plain	Gradually rising slope		Steep slope	Plateau
Soil	Fertile alluvial sandy loam			Shallow with some rock	Very rocky		
Vegetation	Cleared	Wetland	Cleared	Eucalypt forest and some pine plantations		Wetland	Scrub
Transport	Major highway and town roads	Minor roads	Farm roads and tracks	One main access road			Walking tracks
Settlement	Built-up area of 20 000 people	Farm houses – more scattered away from Queensville			Sparse except for village		National park
Landuse	Urban	Intensive farming – orchards, vineyards, vegetables	Extensive farming – cattle and sheep		Forestry		National park

True or false?

6 State whether each of the following statements about the transect above is true or false. For those that are false, rewrite them so they are true.

Statement about the transect	T or F	Correct statement
The land closest to King River has rich soils and has been cleared for houses and farms.		
The soil at Point J is very rocky and the forest is thick there.		
Point K would be suitable for rock climbing.		
There is a national park at Point L, which is also a busy transport hub.		
The area between Points M and N has fertile soils and irrigation so it is not suitable for vegetable, fruit and dairy farming.		
Most trees do not grow above Point O.		

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13 MANUFACTURING FLOWS

Icecream is something that you probably eat without thinking about how it is made. We see steel being used in buildings, bridges and cars, but how is it made into different types of steel? These two everyday items have to undergo a number of stages to convert them from raw materials into finished products. They are manufactured products. Manufacturing is a process that can be best illustrated by a flow diagram because a path can be tracked.

DEFINITIONS

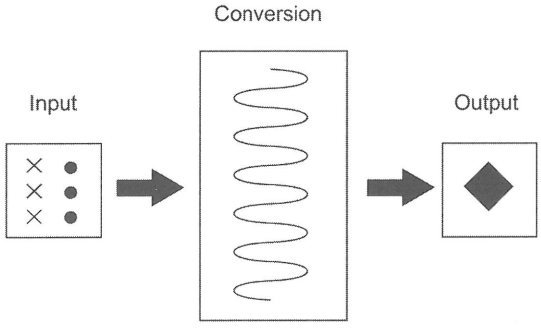
to **convert** is to change something from one condition to another

raw materials are resources – often from the natural environment – before they are made into something else. They are inputs

manufactured products are items that have been made through a number of stages, usually in a factory. They are outputs

a **process** involves a number of stages, from the start to the end

1 Look at this simple flow chart and answer the questions. (Hint: you don't have to use words for all the answers.)



a What is the input?

_____

b What is happening in the middle? _____

c What is the output? _____

d How are the items represented?

_____

e How is the process shown?

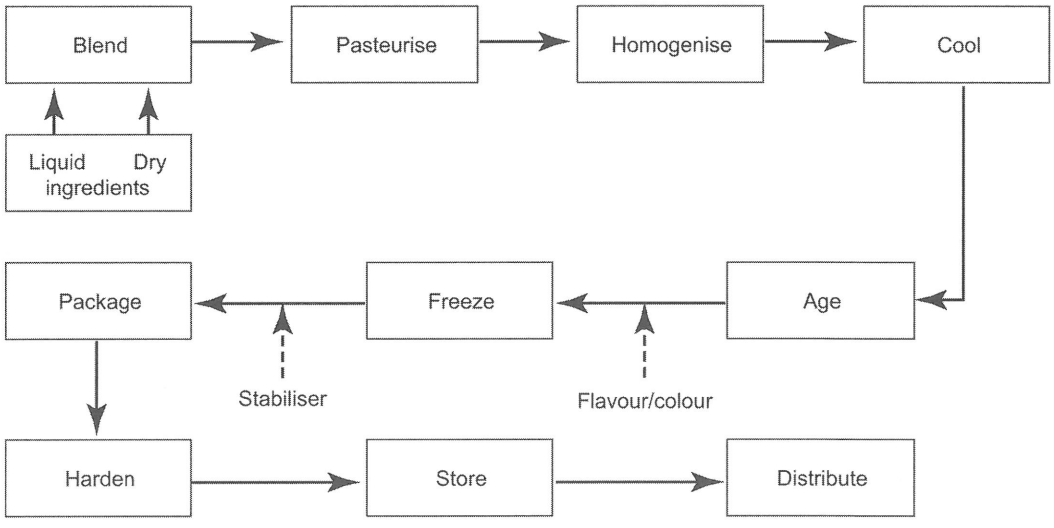
_____

f In a manufacturing process what are the general names for the inputs and the outputs?

_____

_____

2 This flow diagram shows a manufacturing process for an everyday item. Answer the questions about it on this and the following page.



a What are the two main types of input?

_____

b At what stage is the product 'finished' and what is it? (Hint: one of the inputs is milk.)

_____

c How many stages are there in the conversion process? (Hint: storage and distribution are not part of conversion.)

_____

_____

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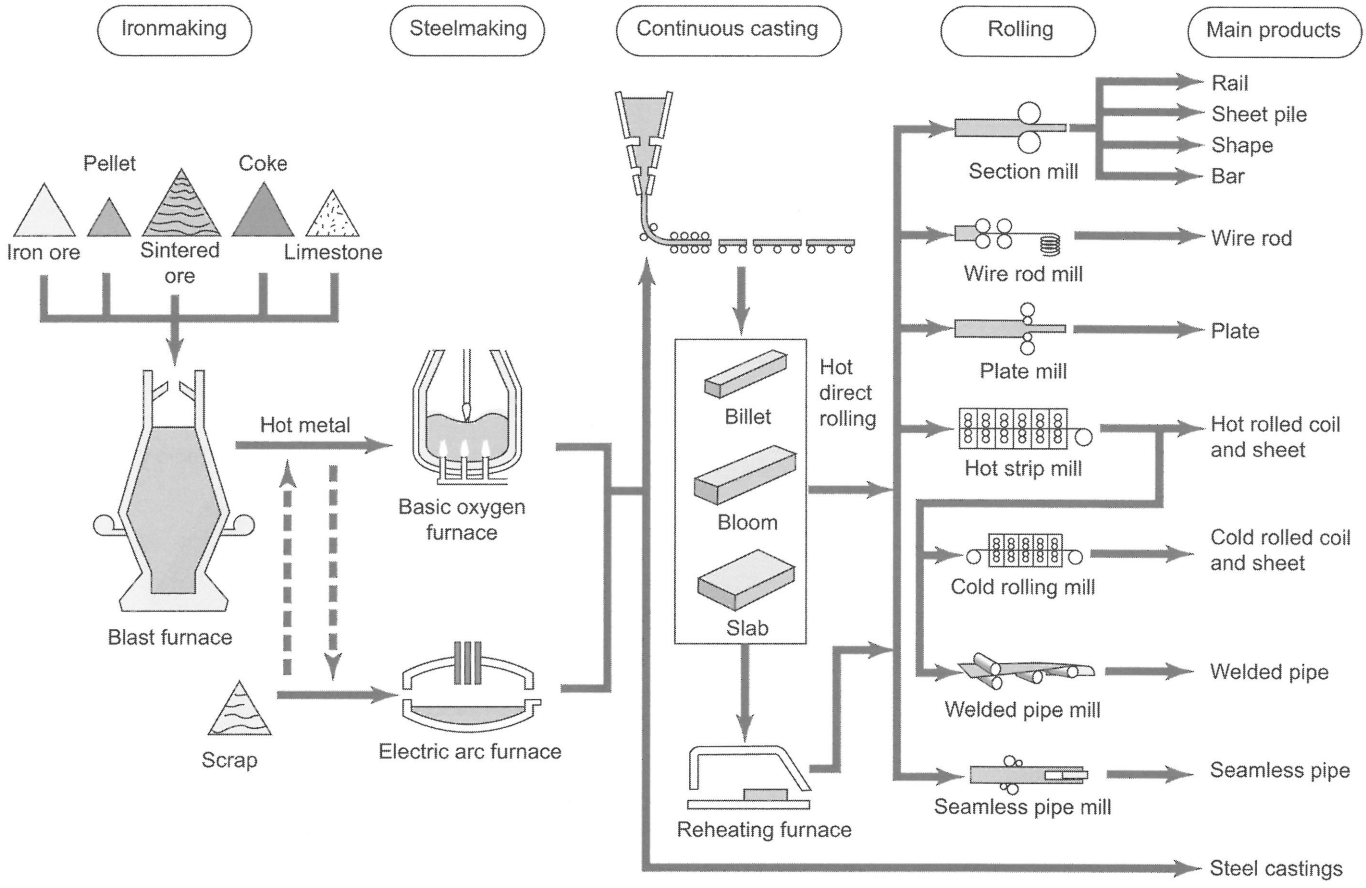
3 As well as the arrows to show the direction of the flow, what other way are arrows used?

4 What is the main technique for displaying the information? _____

5 What way/s could the diagram be enhanced? _____

6 Colour-code the three parts to this manufacturing process – inputs, conversion, and outputs.

7 See how good you are at understanding a more complex flow diagram and manufacturing process. This diagram shows that after basic steel is made it is cast and rolled into a range of products. Use the knowledge you have gained so far to write an extended description of this manufacturing process. Colour-code the diagram into inputs, conversion and outputs. State what techniques are used for the display.



8 **Research**: Choose another manufacturing process and draw a colour-coded flow diagram for it on a separate sheet of paper. (Hint: think of the finished product first. For example a car, bread, a can of drink, canned fruit/vegies, a breakfast cereal, a gold ring, cotton board-shorts, or a washing machine.)

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14 Food chains

Think of your life and how many different things and people you connect with every day. Other people make similar links. Everyone needs energy to make these connections. You get energy from the food you eat. All living things are interconnected. They need to eat food and process it through digestion to release energy.

DEFINITIONS

a **connection** is a link to something or someone

energy is the ability to move and grow

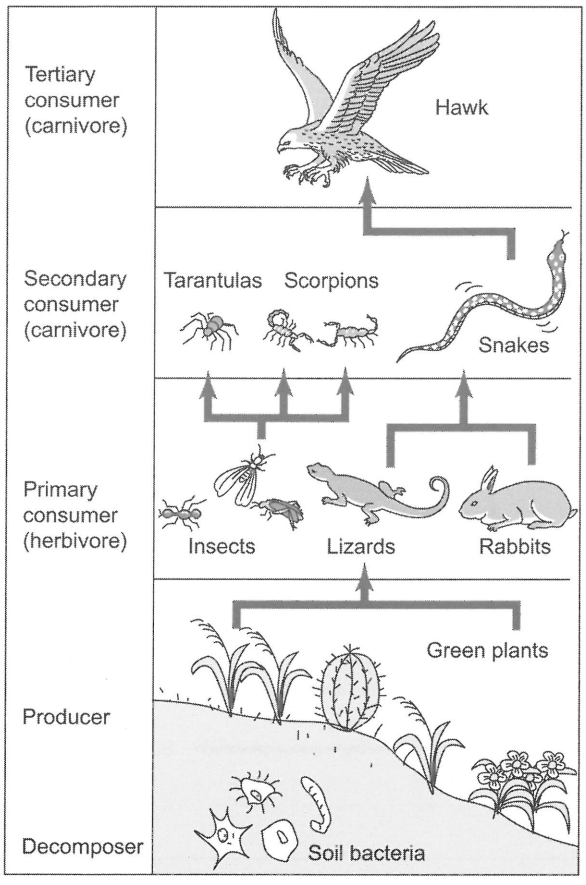
things are **interconnected** when there is a web of links between them

digestion occurs when food is eaten then broken down and absorbed or discarded by the body

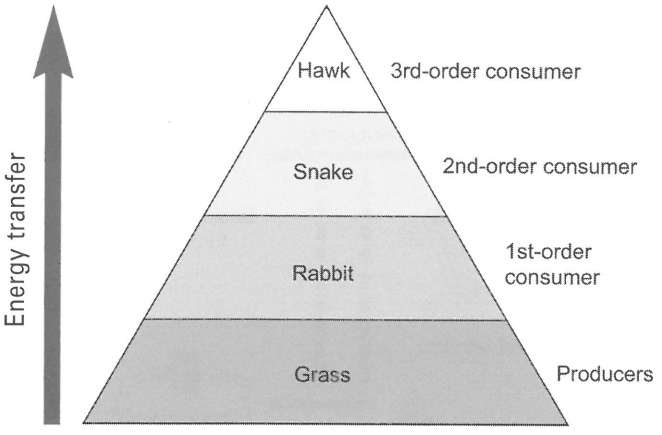
trophic is a level of nutritional energy

1 Examine the two diagrams below then answer True or False to each question **a–g**. Write a correct statement for all the false answers.

A desert food chain



Trophic levels in a food chain energy pyramid



	Statement	T or F?	Correct statement
a	A metal chain has links and so does a food chain.		
b	A secondary producer eats a secondary consumer.		
c	Plants are always at the highest trophic level.		
d	A food web is more complex than a food chain.		
e	If something is harmed in a food chain it stops there.		
f	There is more available energy at the top of a food chain.		
g	Humans are at the top of their food chain.		

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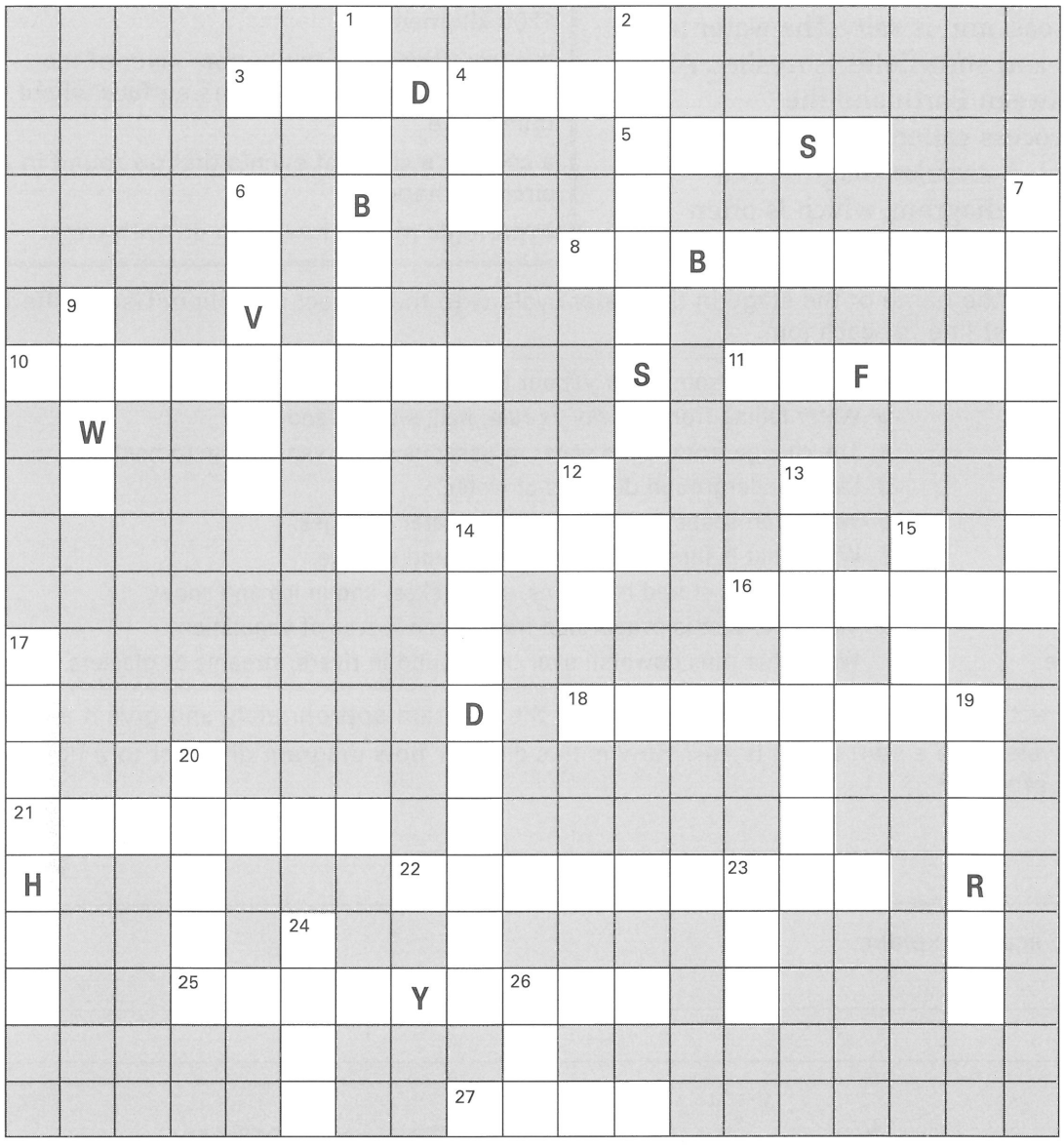
2 What type of diagram is the food chain? Explain. _____

_____

3 Use the two diagrams on the previous page to complete this crossword. (Hint: you will find many of answers to the clues in the labels, but some you will need to work out from the word bank provided.)



Word bank

affect
all
and
carnivore
consumer
consumes
dead
ecosystem
energy
false
food
grass
hawk
herbivore
living
organism
predator
prey
pyramid
secondary
source
tertiary
the
trophic
true
unbalance
web

Across

3. A plant or animal that preys on others and eats them
5. One of the most common producers
8. This can happen if one level of the food chain is harmed
10. A meat-eating animal
11. Organisms are interrelated so what happens to one can_____many
14. A bird, animal or insect that is eaten by a predator
16. The arrow in a food chain shows the direction of _____
18. A longer word for 'eats'
21. The shape that shows there are fewer organisms at each level of a food chain
22. These consumers eat primary consumers
25. A community of living things
27. Decomposers like bacteria and fungi break down _____ things

Down

1. Plant-eating animals
2. A living thing is an _____
4. _____ living things need energy
6. _____ things get energy from food
7. These consumers are at the top of the chain
9. A tertiary consumer in the desert ecosystem
12. The complex relationships between the feeding of animals and plants is a food _____
13. A snake is a secondary _____ in the desert ecosystem
14. Plants are _____
15. True or false: there are food webs in the oceans
17. The name of a level in a food chain
19. The general name for the start of a food chain
20. True or false: humans are not part of a food web
23. A short connecting word
24. Most animals eat more than one type of _____
26. The definite article

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15 THE WATER CYCLE

Water is a natural resource that exists on Earth's surface and in the atmosphere. It can be in a liquid, solid or gaseous state. Most water is held in the oceans and seas, and is salty. The water in rivers, inland lakes and snowfields is fresher. All water circulates between Earth and the troposphere in a process called the water cycle (or hydrologic cycle). A circular diagram is a particular type of flow diagram, which is often used to show this natural system.

DEFINITIONS

the **atmosphere** is the blanket of gases that surrounds Earth to an altitude of over 1200 kilometres

the **troposphere** is the bottom layer of the atmosphere closest to Earth's surface where things live

a **cycle** is a chain of events that go round in a circular shape

hydrologic means having to do with water

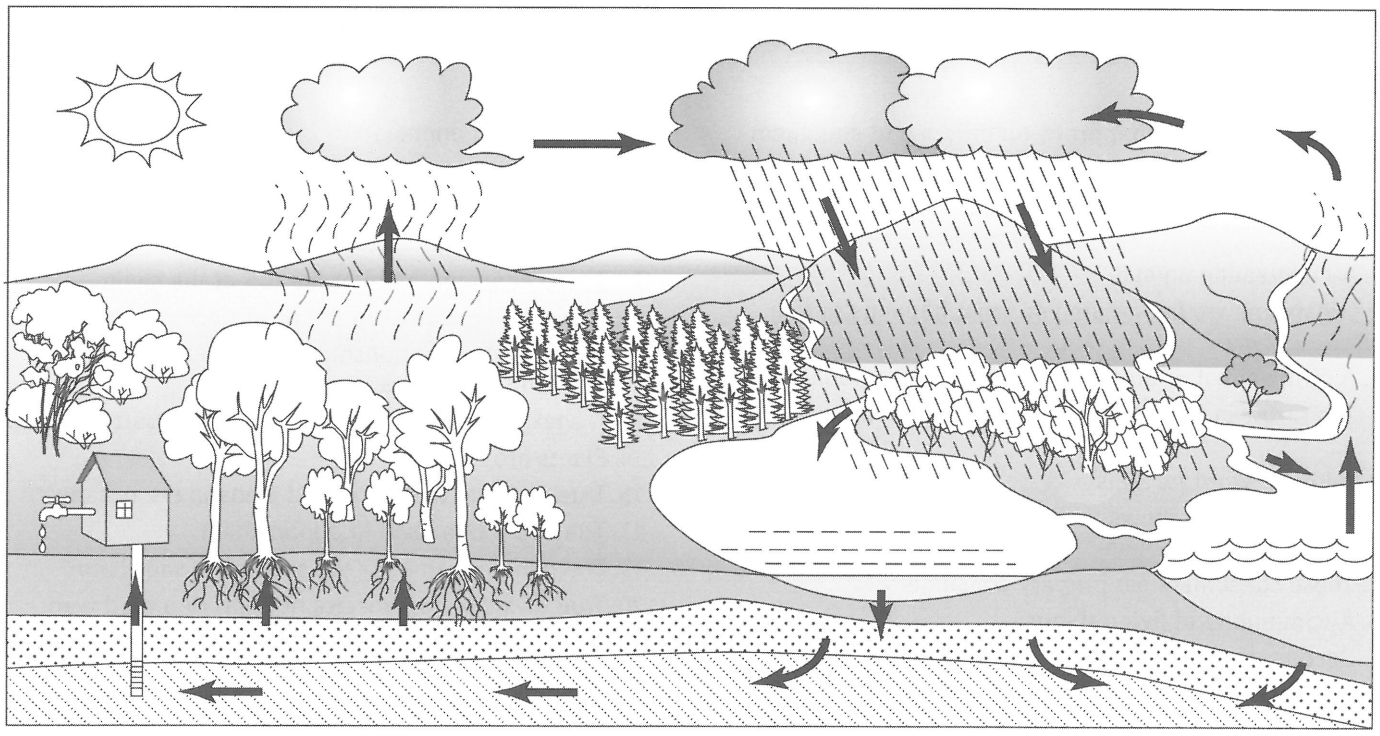
1 In the list below join the name of the stage in the water cycle with the correct definition. Use a different colour or type of straight line for each join.

Evaporation	**a** The change from water vapour to clouds and liquid water droplets due to cooling.
Infiltration	**b** Water falling from the sky as rain, hail, sleet or snow.
Groundwater runoff	**c** The change from liquid water to gaseous water vapour due to heating.
Transpiration	**d** Large underground deposits of water.
Surface runoff	**e** Water that seeps underground into water storages.
Collection	**f** Water that is filtered through the ground surface.
Condensation	**g** Water that is stored in oceans, seas, lakes, and in ice and snow.
Precipitation	**h** The water that is evaporated from green leaves of vegetation.
Groundwater storage	**i** Water that runs downhill over the ground in rivers, streams or glaciers.

2 Now add the correct labels to the diagram below. Colour the diagram appropriately and give it a title.

3 Does the water cycle have a start and a finish? How is this circular flow diagram different to a flow diagram of a manufacturing process?

4 Is the diagram to scale? Explain.



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5 Examine the diagram below. Match the correct labels A–F with the numbered boxes.

Number	Labels
	A Acid-polluted air blown by the wind, sometimes hundreds of kilometres
	B Acid rain falls
	C Wetland plants and wildlife become contaminated and can die
	D Pollutants cause chemical change to water vapour, producing acids, particularly sulphur dioxide and nitrogen oxide
	E Smoke emissions contain acid water vapour
	F Power stations, factories and motor vehicles release chemicals

6 How is acid rain an example of human impact on the water cycle?

_____

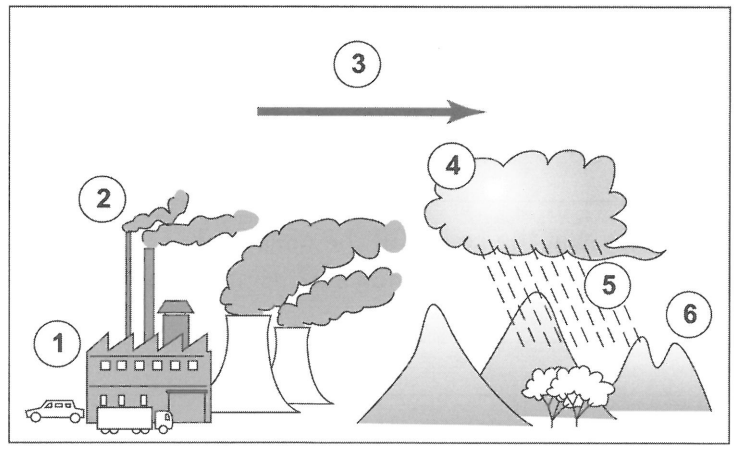
_____

_____

_____

_____

Formation of acid rain



7 Using all of the information in the text and diagrams, complete the passage below. Use labels from the diagrams as well as other words from the text sections.

acid, condensation, contaminated, continuous, cyclic, downhill, end, evaporates, hail, hundreds, natural, oceans, plants, polluted, precipitation, rivers, runoff, snow, trapped, troposphere, water cycle

Water is an important _____ resource. It is found on Earth's surface and in the _____, which is the layer of the atmosphere closest to Earth. Geographers call the circulation of water between Earth and the atmosphere, the _____ _____.

There is no real start or_____ to this process as it is _____.

At the Earth's surface water is stored naturally in lakes and the _____. Water is also part of plants. When water is heated it _____ from water storages and transpires from _____. The rising air cools and _____ occurs. Clouds form and when the air becomes colder they cannot hold all the water droplets.

_____ can fall in the form of rain, _____, _____ or sleet. When precipitation reaches Earth, some of it runs _____ over the ground in streams, _____, and glaciers. Some infiltrates the ground. Some of the groundwater is _____ in underground cavities in the layers of rock, and some becomes groundwater _____, which travels slowly to collection points such as lakes and oceans.

Because the water cycle is a _____ process, a problem in one part can be transferred to another. _____ rain is an example of this because _____ air can be carried _____ of kilometres from its source and cause plants and animals to become _____.

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16 THE POVERTY CYCLE

It would be helpful for students to use the global statistics pages of their atlas to complete question 6.

In the natural environment there are geographic events that happen in a cycle, such as the water cycle, the carbon cycle and the seasons. Cycles can also be found in Human Geography. Poverty is one example. In some countries of the world people find it hard to break out of the poverty cycle. The cycle passes from one generation to the next. Poverty causes more poverty. The cycle of disease is similar. It is often those in the poverty cycle that are also badly affected by the cycle of disease. Both cycles affect more people in developing countries compared to developed.

DEFINITIONS

Human Geography is the study of how different communities and cultures interact with each other and the natural environment
a **cycle** is a chain of events that go round in a circular pattern
poverty is the situation of having little money and only a few basic possessions
a **disease** is a bad illness
a **developing country** has much lower everyday living conditions than a developed country
a **developed country** has good everyday living conditions for most of the people

1 The diagram below shows a continuous cycle. In general terms, what does this mean?

_____

2 What is at the centre of the diagram? _____

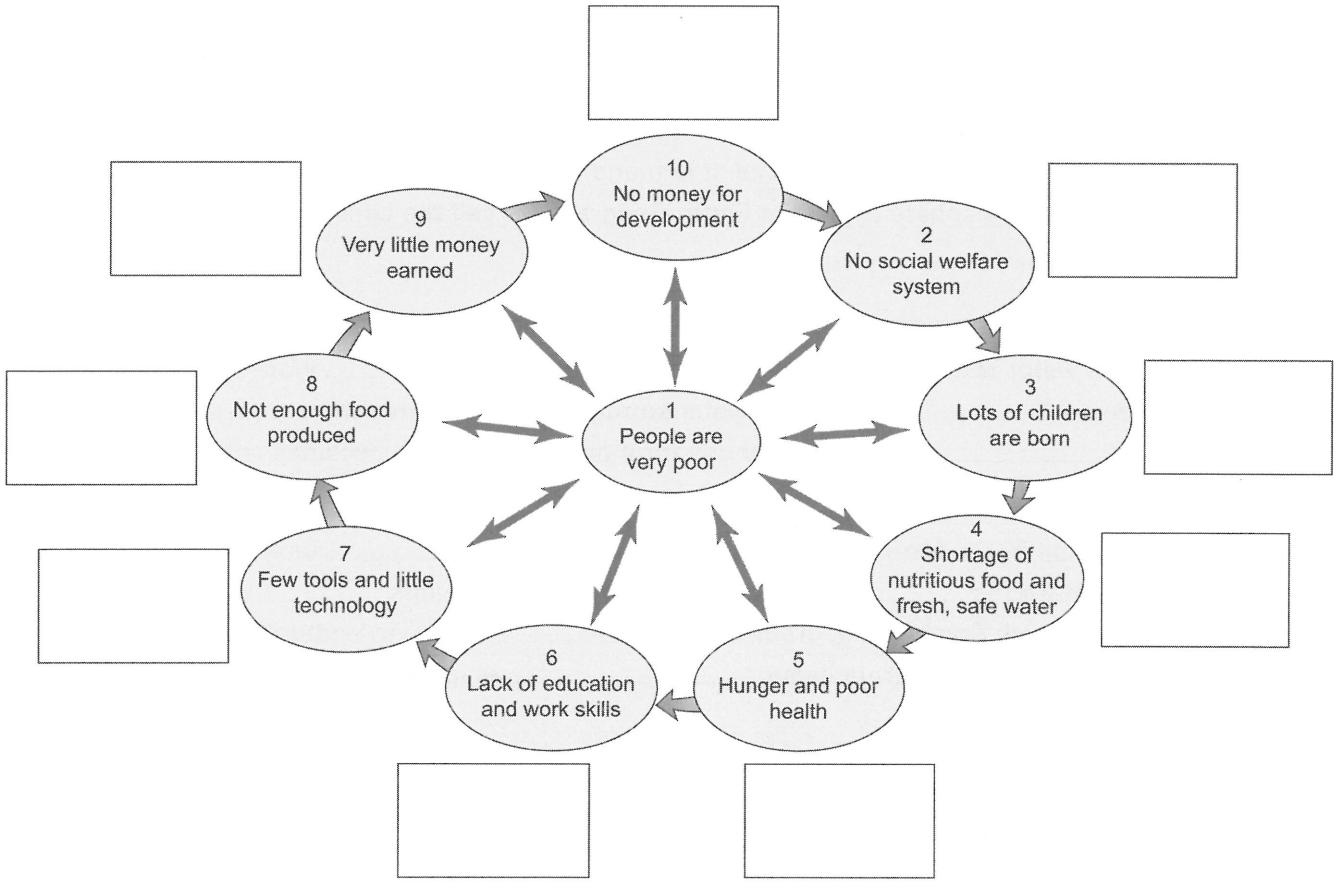
3 What do the arrows indicate? (Hint: there are two levels of arrows, so describe both.)

_____

_____

4 Next to each stage, write another label in each rectangular box that could be a possible solution.

5 Colour the whole diagram, including your new labels, and provide a key. Add a title. (Hint: the colour for the solutions should be different to the inner cycle.)



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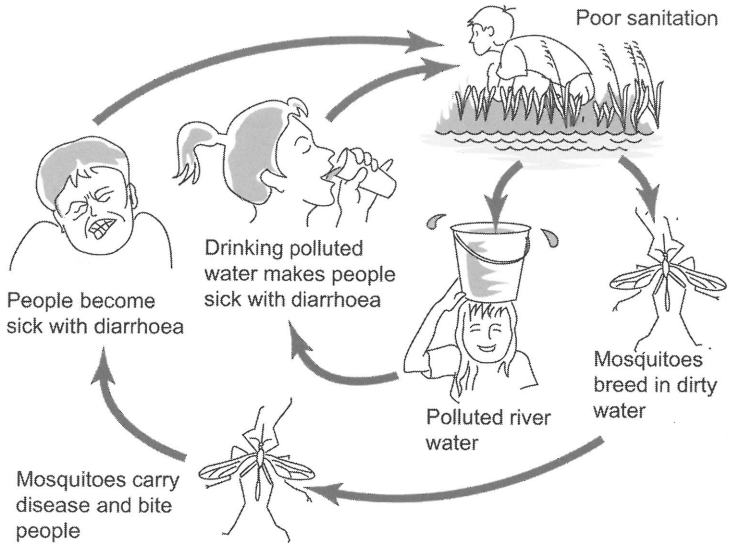
6 Complete the table below to distinguish between developing and developed countries. (Hint: use an annual income (GDP) per capita of US$15 000 as a cut-off between the two categories.)

Global region	Developing country	Developed country
Africa		
Asia		
Europe		
North America (north of Panama)		
Pacific (Oceania)		
South America (south of Panama)		

7 Are there any gaps? If so, what can you conclude?

8 Diarrhoea is the second biggest killer of children in the world. To the right of the diagram below describe the cycle shown. Add a title.



9 Both diagrams in this topic can be classified as circular. How do they differ to straight-line flow diagrams?

10 True of false? Both of these cycles only occur in the poorer countries of the world. Explain your choice.

11 Another example of a cycle is the recycling of aluminium cans. On a separate sheet illustrate this in a diagram. Add small graphic symbols and colour to enhance your diagram. The steps are:

1. Aluminium cans are collected as part of kerbside collections and delivered to a Materials Recovery factory.	2. The aluminium is separated from the other recyclables either by hand or using a 'Eddy Current' Separation unit.
3. The cans are weighed and squashed together and baled into blocks. These blocks are taken to the furnace.	4. The blocks are tipped into a rotary furnace where they are melted.
5. The molten aluminium is poured into a mould to make ingots.	6. The ingots are taken to a factory where they can be formed into new aluminium items.

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17 SEASONAL CYCLES

One of the most obvious geographic events that occurs on an annual cycle is seasonal change. Many human activities over thousands of years have been tied to the seasonal calendar, especially those relating to living off the land. Even with today's technology, many farming, fishing and forestry activities depend on the seasons. In the natural environment, the migrations of birds and whales are two examples of seasonal cycles. One of the best ways to show seasonal change is by a circular diagram.

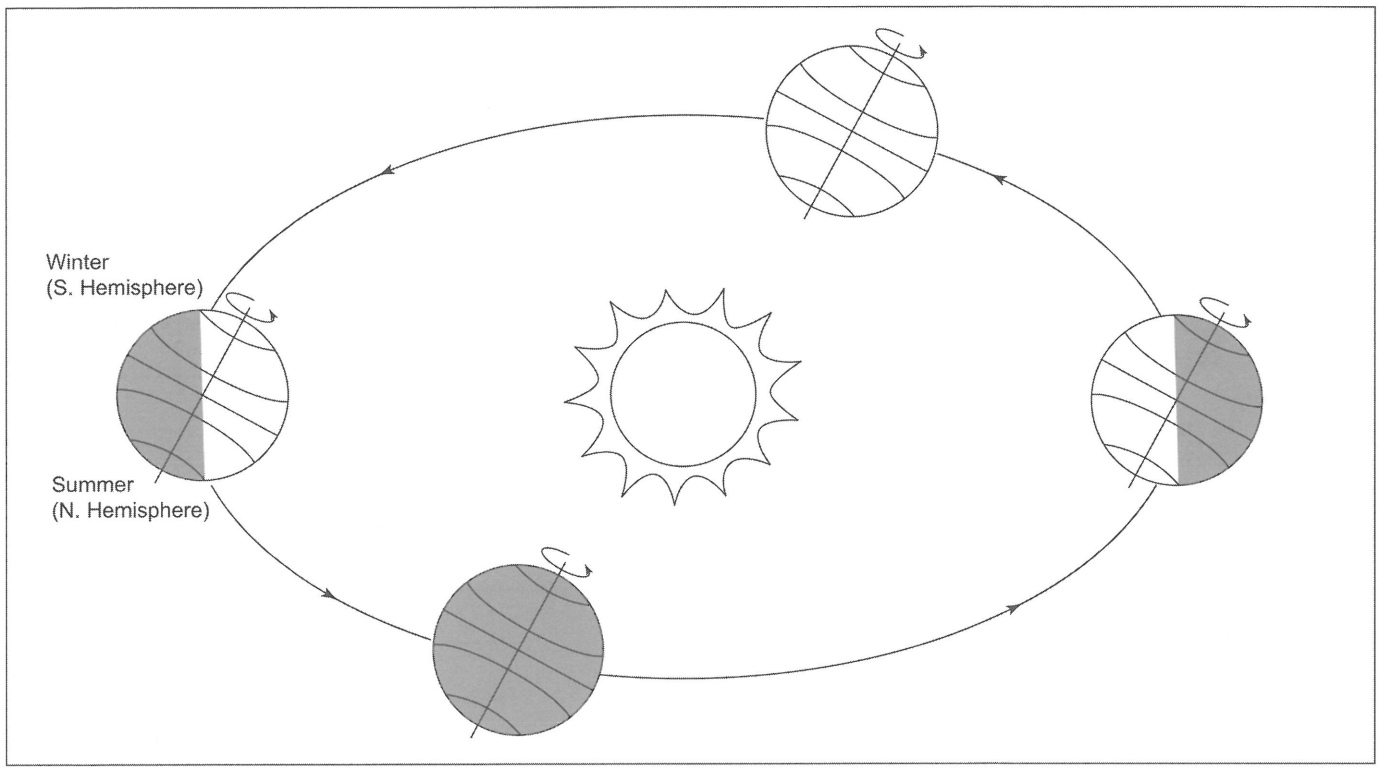
DEFINITIONS

seasonal means relating to one of the four seasons – spring, summer, autumn or winter

technology is the use of computers, machinery and scientific research to help improve things

migration of animals is the seasonal movement from one place to another for feeding and breeding

an **orbit** is the circular, or near circular, path a planet makes around the sun



1 In the diagram above, what does the circular arrow over the top of Earth show?

_____

2 What cycle does this movement cause? _____

3 What important other cycle does the diagram show? Describe what happens. (Hint: use the word 'orbit' in your answer.)

_____

_____

_____

_____

4 Add the missing labels to the two globes. Write the month for the time of year next to each of the four globes. Give the diagram a title.

5 Is the diagram drawn to an accurate scale? Explain. _____

_____

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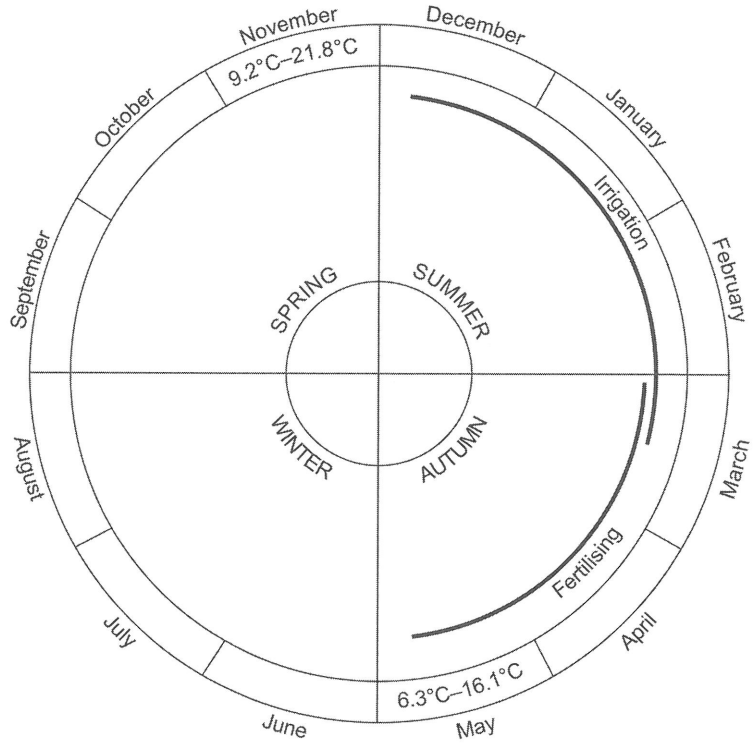
6 The table below has been compiled from the website of a well-known winemaker in Victoria's Yarra Valley – Domaine Chandon at <domainechandon.com.au> – and from climate information from the Bureau of Meteorology. It describes what happens in the vineyard before the winemaking begins. Use the information to complete the outline circular diagram. Place the things that happen to the grapes in each quarter then the six main activities in arcs under the appropriate months. Follow the example to insert the average temperatures for each month.

Season	In the vineyard	Average daily max. and min. temperatures °C
Summer	The fruit growth accelerates and the grapes begin to soften. Sugar content increases, and acidity decreases. Grape colour develops.	Jan 26.3 and 11.0 Feb 26.6 and 11.7 Mar 24.0 and 10.3 Apr 19.0 and 8.5 May 16.1 and 6.3 Jun 12.2 and 4.1 Jul 12.0 and 4.0 Aug 13.6 and 5.1 Sep 16.3 and 6.1 Oct 19.1 an 8.0 Nov 21.8 and 9.2 Dec 23.9 and 10.5
Autumn	Picking starts early in the season. Picked grapes are sent to the winery.	
Winter	Preparation for next season starts. Machines remove the bulk of the previous season's growth, then spur pruning is done by hand. Buds appear at the end of the season.	
Spring	Buds open as the temperature rises. Young vine shoots emerge. Flowers develop and are pollinated by bees. Tiny bunches of grapes appear. Shoots attach to the wire trellising.	

Activities:

- irrigation from December through to March
- pesticides and herbicides in Spring and Autumn
- repairs and maintenance of trellising in Winter
- fertilising in Autumn
- weeding in Winter, Spring and Autumn
- pruning in late Autumn and early Winter

7 Use appropriate colours to enhance the various elements of the seasonal cycle in the vineyard at Domaine Chandon. Include a key if you think this will add to the meaning of your diagram. Also add a large label in the centre in all capital letters that is the title to the diagram.



8 This diagram is a type of timeline. But most timelines are linear. Explain the difference between the two different types of representation.

_____

_____

9 On a separate sheet of paper, create at least 6 questions related to your completed diagram. Also provide the answers.

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18 SIMILARITIES AND DIFFERENCES

A Venn diagram uses circular areas to organise ideas and to show things that groups have in common. The diagram consists of two or more circles, each representing a specific group. Any values that belong to more than one set are placed in the sections where the circles overlap. This overlap area is of most interest as it shows what sharing of conditions occurs. Venn diagrams are useful tools for geographers who have to solve problems. This way of showing logical relationships is named after the mathematician John Venn who first used these diagrams in the late 1800s.

This topic is suited to small group activities. Blank outline Venn diagrams are provided on the inside front and back covers.

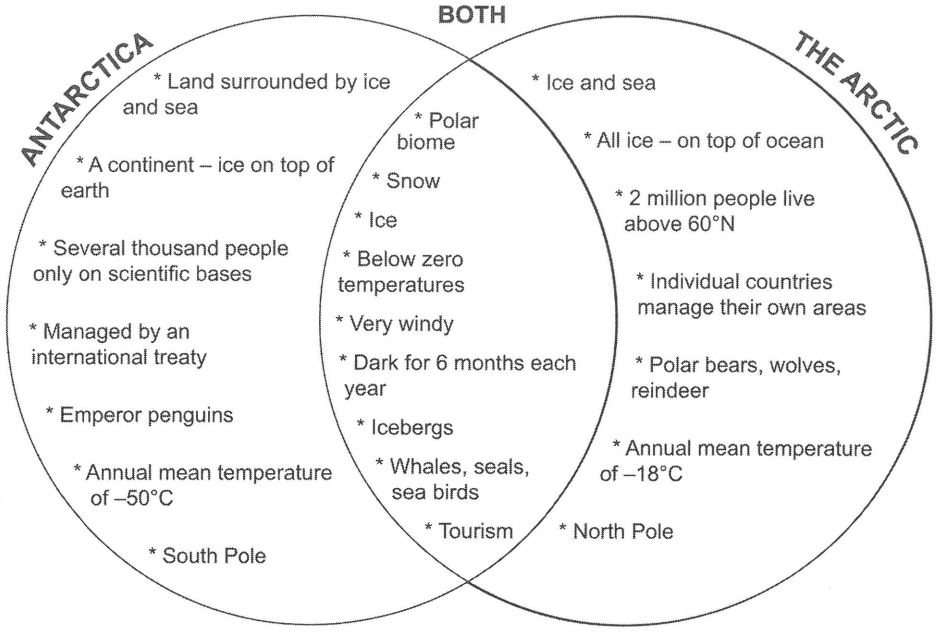
DEFINITIONS

logical relationships are connections that groups have in common

values are features, functions or measurements

a **set** is a group or collection of similar things

the **CBD** of a large city is the central business district



1 What comparison is being made in the Venn diagram above?

2 List the things about the climate and physical environment that both regions have in common.

3 List the following about the wildlife:

a types they have in common: _____

b found only in Antarctica: _____

c found only in the Arctic: _____

4 Which place is a separate continent? Explain. _____

5 Where are you likely to live if you are an indigenous Innuit or Lapp? Explain.

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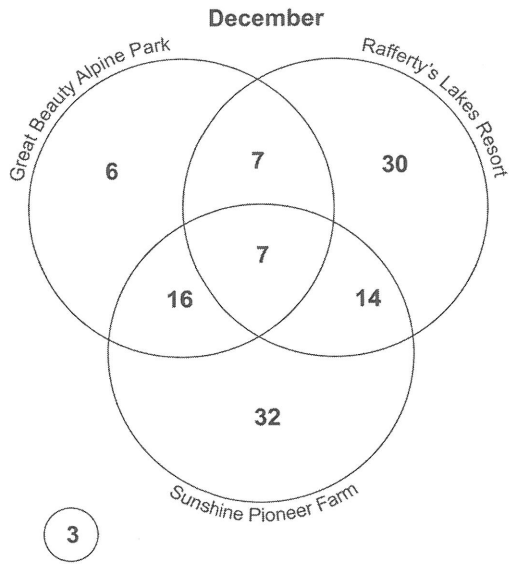
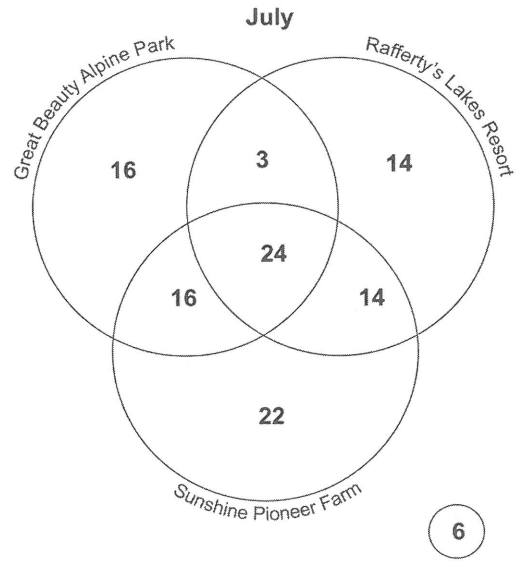
6 Three different situations can also be represented by a Venn diagram, such as the ones below, which were compiled from surveying a group of students in Adelaide about three possible destinations for a school camp at two different times of the year. The students could make either 1, 2 or 3 choices.





a How many wanted to go to the Great Beauty Alpine Park in summer? _____ in winter? _____

b How many wanted to go to Rafferty's Lakes Resort in summer? _____ in winter? _____

c How many wanted to go to Sunshine Pioneer Farm in summer? _____ in winter? _____

d What do the figures in the small circles mean?

_____

_____

e How many students were surveyed? _____

f What problem or problems did the teacher have to solve? _____

_____

g What decision/s do you think the teacher made?

_____

h How did the Venn diagrams help the teacher's decision-making?

_____

Compile your own Venn diagrams

7 Make a comparison of your own local area – within a diameter of about 5 kilometres from your home – and the CBD of the capital city in your state. Use the planning grid on the 2-circle Venn diagram template to list the elements of each environment, then those that they share. Insert these into the appropriate parts of the circles. Write a short description about what the diagram tells you. Your problem is: Where would you prefer to live? Enter this, then add your solution.

8 There is a problem that the local council needs to solve because a large factory has been demolished and the council has bought the land. There are different views about how it should be used, so the council took a survey to find out which of three options was the most popular: a car park, parkland, or a child-care centre. Create the Venn diagram on the 3-circle template. Write the number of people surveyed and the decision you think the council reached.

- 250 people want a car park
- 100 people want parkland
- 100 people want a child-care centre
- 50 people want both a car park and parkland
- 30 people want a skateboard path.
- 400 people want both parkland and a child-care centre
- 10 people want both a car park and a child-care centre
- 150 people want the site divided into all three options

9 Create a Venn diagram of your own that solves a geographic problem or makes a comparison between geographic features.

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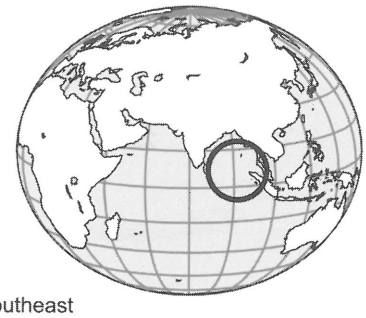
19 MIND MAPS SHOW LINKS

This topic is suited to small group or class brainstorming activities.

A mind map is not a typical map found in an atlas or a street directory. It is a diagram that shows the links between a main idea or theme and its subsets. A mind map provides an overview or summary that is easy to interpret because it is in graphic form. A typical mind map can look like a tree or river with lots of branches and sub-branches. To create a mind map, start with a main theme or idea and group subsets around it, creating branches for each new subset or sub-subset. Some mind maps use colour, small pictures or photos to add interest.

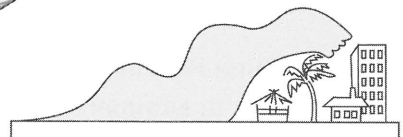


Indian Ocean, Southeast Asia — WHERE?

A giant wave — WHAT?

26 December 2004 — WHEN?

Resulted from undersea earthquake of 9.3 off northern Sumatra — WHY?



THE 2004 TSUNAMI

no warning

IMMEDIATE IMPACT

- panic and chaos
- death and destruction in 11 countries
- entire coastal areas wiped out in 6 main areas
- total death toll of more than 230 000
- huge global relief program
- loss of tourism

the future?

LONG-TERM IMPACT

- coastal dwellers live in fear
- rebuilding will take a long time
- women and children numbers decimated
- environment harmed
- 1.6 million displaced people

1 What is the central theme of the mind map above? _____

2 Without reading all the details, what impression does it give you at first glance? How do the small graphics help?

_____

_____

3 What is the main subset of information on the left side? _____

4 What is the main subset on the right side? _____

5 State what the two action connectors say and why they are an important part of the mind map.

_____

_____

6 Add another subset that runs vertically from the main theme. Use the connector 'Implications for research'. Then add these notes:

- plate tectonics in Southeast Asian region
- tsunami warning system for Indian Ocean
- establishing livelihoods away from coastal strips

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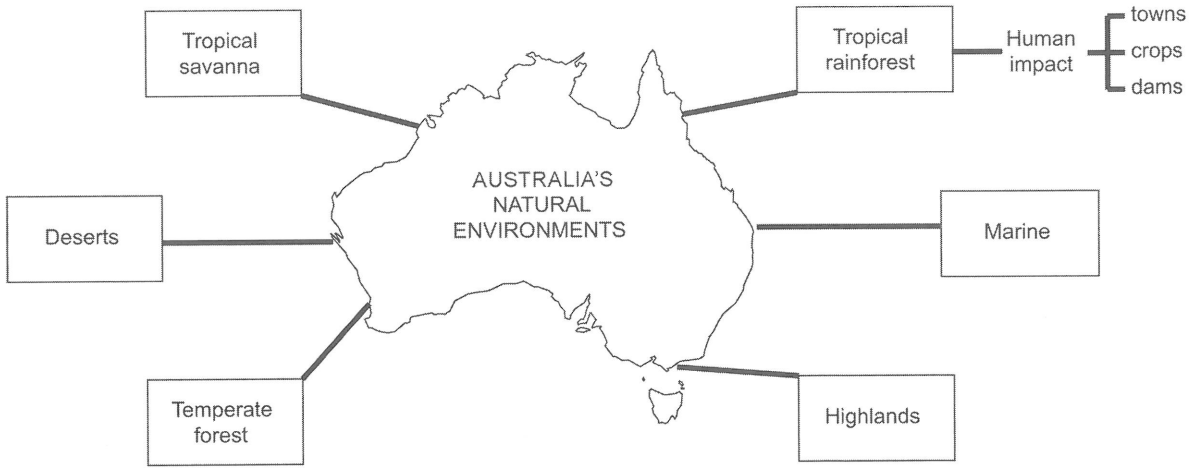

7 Rectangular boxes, circles or ellipses are often used in mind maps to frame each main idea. Further away from the central theme, the frames might be dropped and just the label written. The mind map above has been started for you. Add as many ideas to it that you can about human impact on each environment.

8 Choose four shades and colour the main divisions of the mind map – central theme, subset 1, subset 2, and subset 3. Describe what effect colouring the map has.

_____

_____

Construct a mind or concept map

Mind maps provide a good opportunity for a group to brainstorm the ideas. Detailed mind maps are also called 'concept maps' because they build on a few main concepts. The links are labelled and sometimes the direction is shown by arrows on the connecting lines.

9 In small groups follow the steps below to create a poster-size mind map that can be displayed in class. It would be a good idea to have a set time limit, say 30 minutes, for the completion of the whole activity. Choose one of these four topics as the central theme:

Tropical Rainforests **Global Tourism** **Natural Disasters** **Careers in Geography**

Step	Process
1 Schedule	Once you know how much total time you have, allocate a time span to each of the steps below. Write the minutes next to the step.
2 Brainstorm	From memory write down on small pieces of paper or sticky tags as many words or short phrases your group can associate with the topic. They should be ideas not actions. Don't discuss the importance of anything at this stage.
3 Organise	Spread out all the pieces of paper from step 2 so that the whole group can see them. Try to find some common threads for several bits of paper and bunch these together. Identify terms that each bunch could be grouped under. Write these on a different-coloured piece of paper and add to the top.
4 Layout	On your poster paper draw a box around the central theme. Then try to visualise the best layout for the groups you have categorised in step 3 by placing the bits of paper onto the poster. Rearrange them if necessary.
5 Link	Think about lines and arrows that could be drawn to show some action or relationship between items. Then create a word or short phrase as a link.
6 Complete	Once the group has agreed on the arrangement of all the items, convert the map into a permanent form by drawing and labelling the concepts and links. Use colour, different label fonts/strengths and even small graphics to better communicate your group's understanding of the central theme.

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20 SKETCHING HUMAN IMPACT

Everywhere you look you can see the impact of humans on the natural environment. Just think of a large city, a huge shopping complex or a marina and high-rise apartments at a seaside resort area. The built environment dominates the natural environment in many places.
Geographers often sketch what they see around them. Sketch diagrams are a tool that can show human impact on a landscape. Field sketches help remind you of the conditions. They are like any other diagram in that they should be clearly labelled and have a title.

DEFINITIONS

an **impact** is a powerful effect

the **natural environment** is all living and non-living things that have been formed by nature

the **built environment** is what people have made or grown

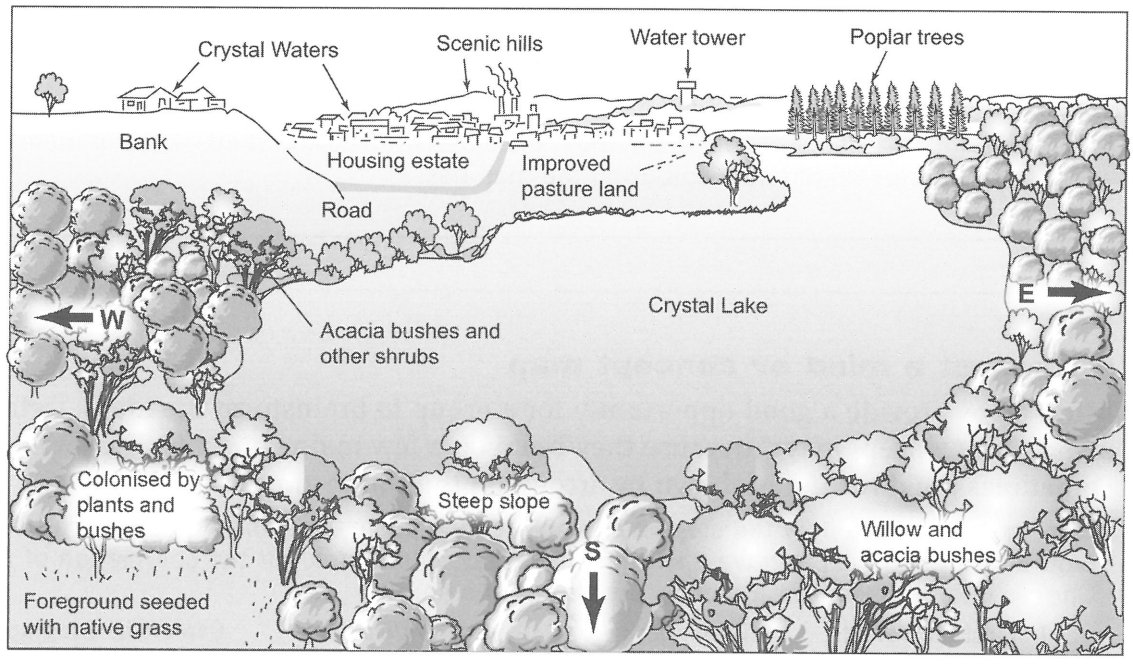
to **dominate** is to overshadow or wipe out

to **sketch** is to draw quickly and without too much detail or accuracy

fieldwork is when a geographer goes out and about to observe and collect data

1 This diagram is an example of a good field sketch. The table below lists its good qualities. Add a number to the sketch to identify each quality and write the corresponding number in the table. Colour the sketch appropriately, as colour is another important quality of a good field sketch.

Field sketch of Crystal Lake and surrounds, drawn from the south point approximately 5 kilometres from Scenic Hills estate, by S Morrison, 19 December 2005, 11.30 am, fine weather



Number	Quality of a good field sketch
	Shape of the land and major landforms
	Major built features
	Major transport routes
	Major natural features
	Short notes about areas being studied
	A border
	A note of the date and time
	An approximate scale
	Direction pointers
	Any specific names of known features
	The location of the sketcher
	A note about the weather conditions

2 In your own words describe what this field sketch tells you about the environment that was observed.

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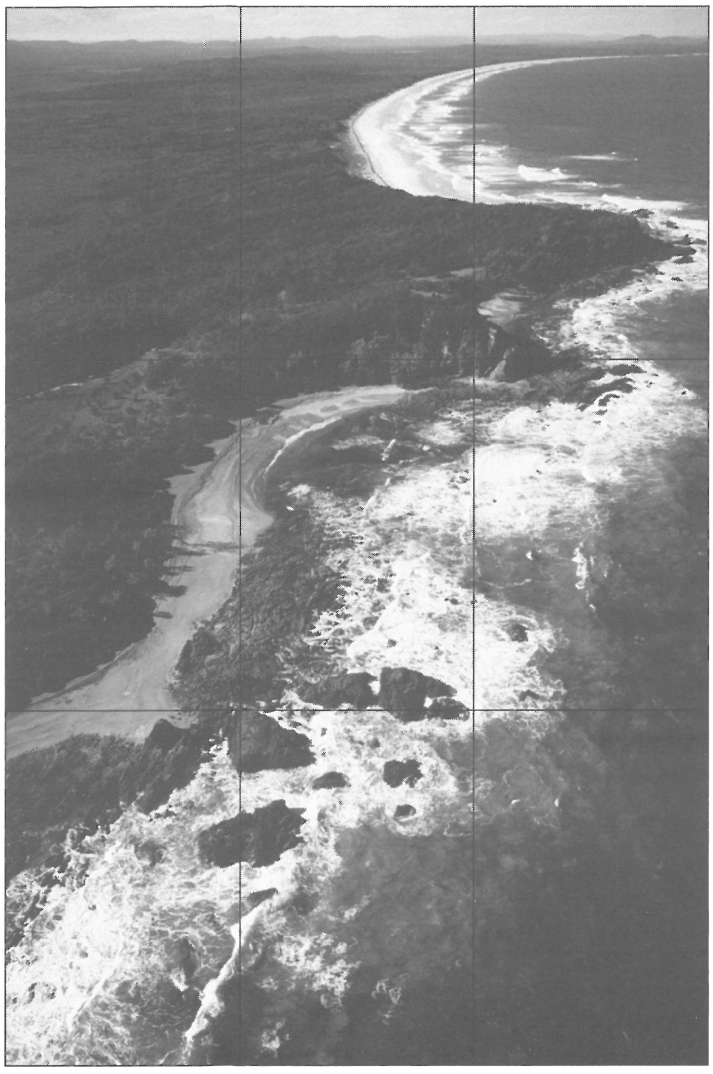
3 You can practise drawing sketches from photos. Follow the step-by-step process to build up your final diagram in the space next to the photo.



Step 1	This has already been started for you. The photo has been divided into background, middle ground, foreground, and left, centre and right. Copy this grid into the boundary on the right as a guide.
Step 2	Observe the outline of the physical landscape first and draw this. Focus on one piece of the grid at a time and gradually build up the shape.
Step 3	Add some parts of the detail. Choose the main features.
Step 4	Label the main features. Use general geographic names and these real names: Macquarie Ranges (mountains in background); Sunshine Beach (backgound); Clear Waters National Park (to left of beach and cliffs); access road; and Vision Headland (middle ground).
Step 5	Add the other details to your sketch and lightly shade it. Details: This photo was taken on 19 December 2005 looking north to Sunrise Beach. The distance along the shoreline from foreground to background is approximately 8 kilometres.

4 A large resort is planned for the southern shore of Sunshine Beach. Stage 1 will consist of a 3-storey hotel 50 metres from the sand, a general store and petrol station just north of the hotel, and a caravan park on Vision Head. The access road will also need upgrading to a 2-lane bitumen road. Add these to your field sketch and say how the diagram helps you identify human impact on coasts.

5 How could field sketches be used to show change over time?

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21 Structure of organisations

Most people live in households as part of a local community. If they go to school, work, church or hospital they become part of other groups for some of the time. These groups are organisations that have their own special functions. Each organisation has a structure that can be shown in a diagram. An organisation chart shows the reader at a glance all the different components of an organisation *or* of an organised process. It is similar to a flow chart in that energy (in the form of power) flows from one point to others.

DEFINITIONS

to **organise** is to get people together for a particular purpose *or* to arrange something

an **organisation** is a group of people and functions organised to do some type of work or activity

organisation is a planned arrangement

a **function** is a purpose or the main thing that something or someone does

the **structure** of something is all the parts that it is made up of

1 The list below shows a simplified structure for a secondary school. Think about your school as an organisation. In column 2 estimate the number of people at each level. In column 3 change the numbers into a box symbol ☐ to represent the numbers at each level. The first row has been done as a guide. Place the boxes in the centre of the column and work out from the centre evenly to the right and left. If there is not enough space within the column, only fill the space. Don't go over the borders. (Hint: your symbols should all be the same size, so use 2 rows of symbols to fill the students' row.) Shade each row a different colour.

Function	Number of persons	Simple diagram
Principal	1	☐
Deputy or Vice Principal		
Curriculum Coordinator		
Faculty Head or Head Teacher		
Teachers		
Students		

2 What observations can you make about the pattern of symbols? What does this mean?

_____

_____

3 What type of diagram does your pattern resemble? _____

4 There are thousands of different organisations in the world, each with its own function and structure. Knowing about some of these organisations is important in the study of Geography. There are organisations at the local or state level, national or country level, and the global level. Some examples are given in the table below. Tick the level that you think each belongs to: L/S = Local or State, N = National, G = Global.

Organisation	L/S	N	G	Organisation	L/S	N	G
1 United Nations				9 Australian Government			
2 WA's State Emergency Service				10 International Red Cross			
3 Benalla Landcare Group				11 Kyoto Protocol			
4 Landcare Australia				12 Great Barrier Reef Authority			
5 World Health Organisation				13 CSIRO			
6 Kakadu National Park				14 Tasmanian Parliament			
7 Riverview WaterWatchers				15 World Heritage Committee			
8 Geography Department of the University of Sydney				16 Gold Coast City Council			

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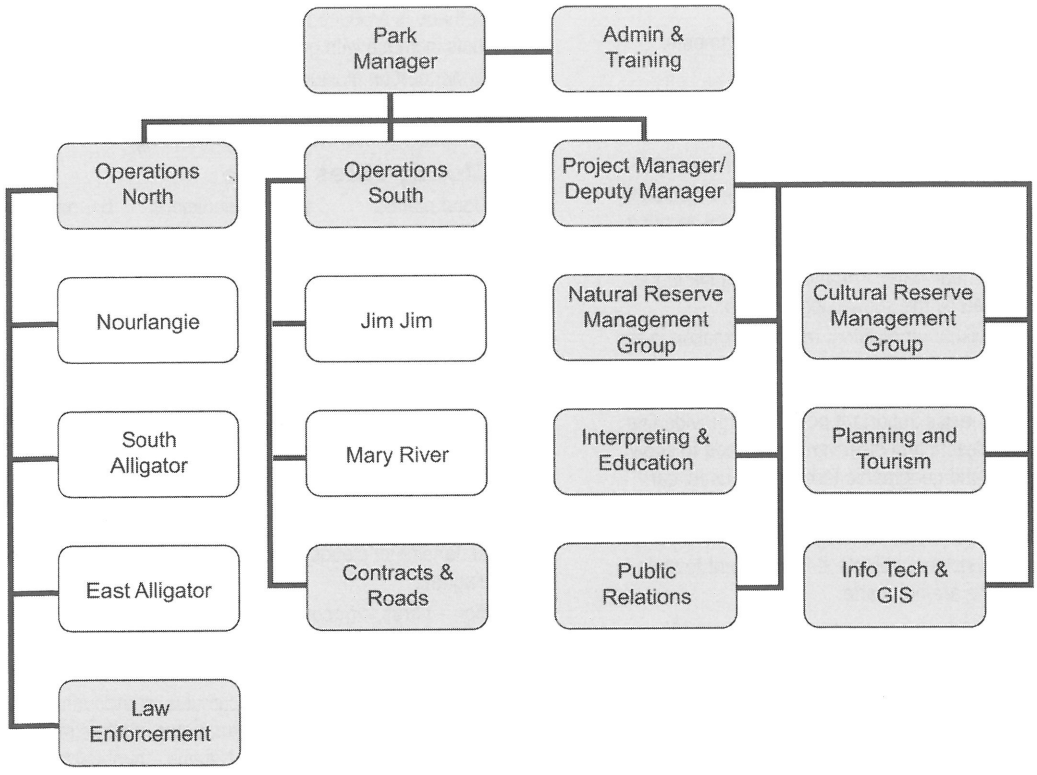
5 The organisation chart below for Kakadu National Park was sourced on 28.04.2005 from the website <http://humanities.cqu.edu.au/geography/GEOG11024/images/KakaduOrg.gif>.



a Colour-code the chart by shading the white boxes a light green and light blue for the grey boxes.

b The green boxes are specific regions of the park. What does this mean? How many are there?

_____

c The blue boxes are the functional units. What does this mean? How many are there?

_____

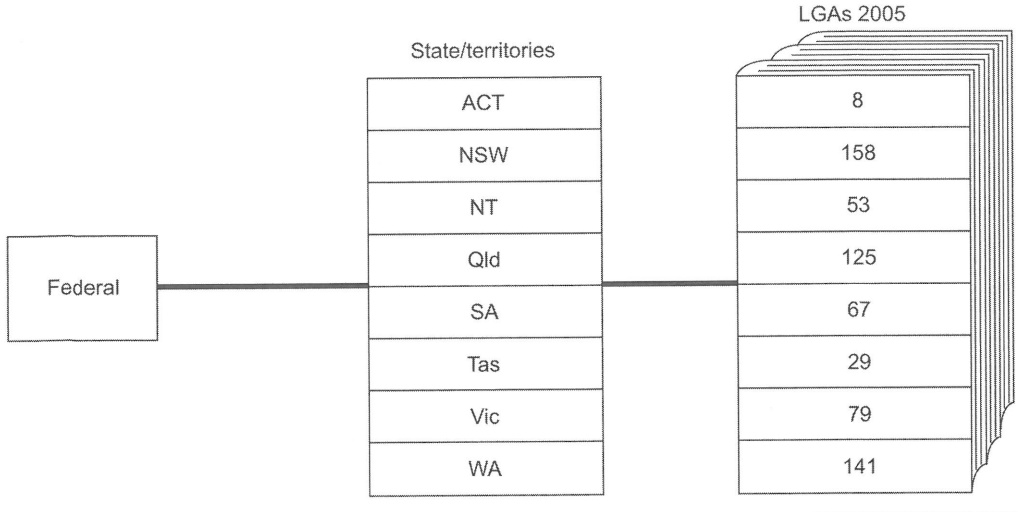
d Add a key to the right side of the diagram.

e Who has direct responsibility for enforcing the law in the park? _____

f How many functions does the Project Manager manage? _____

g If the Park Manager gets ill or is on leave who would take overall responsibility? _____

6 On a separate sheet of paper describe what the organisation chart below shows. What technique is used to show the LGAs? Why? How does this diagram help your understanding?



_____

Note: Local government areas (LGAs) = cities (urban or large regional), municipalities, shires, regions, and in NT, community government councils, and incorporated associations

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Selected Answers

1 Diagrams in Geography

1a Possibilities include: airstrip, sports stadium, houses, apartments, offices, bridge, marina, roads

1b Possibilities include: trees, mountains, river, forest, snow, sand

1c Boating, swimming and driving

1d The general description should indicate that a natural environment of a river or similar water body bordered by a natural forest has been settled by humans and changed to suit housing, water-related, recreational, farming and transport activities. It could emphasise that many people like to live in beautiful surroundings, especially near water. The mountain region in the background gets cold as evidenced by the snow-capped peaks.

2 Diagram advantages: instant visual recognition, more information in the same space. Diagram disadvantages: need to interpret to get the information; usually not to scale; can take a long time to construct. Written description advantages: can emphasise important points; can provide fine detail, such as distance. Written description disadvantages: need to know and understand all the language and geographic terminology used; can become boring.

4 The statement 'Not to scale' means that the diagram does not represent accurately the true distances and sizes of objects. It is important to state this so that misleading conclusions are not made.

5 Student's own interpretation.

7 Depends on student's own textbook.

2 Planet Earth

1 Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto

3 A scaled diagram is drawn accurately in the correct proportion to reality. There are two major problems in drawing our solar system to scale on paper (even at poster size): (i) the size of the sun and the planets is small compared to the vast distances between them (ii) the sun is enormous relative to the sizes of the planets.

6 outer, continental crust, soil, rocks, oceanic, thinner, oceans, mountains, plateaus, plains, sand

3 Above Earth's surface

1 'Sphere' is the suffix. It is used because a sphere is a round shape, and this represents Earth as a globe and the layers of atmosphere surrounding it.

2 Student's own definition.

4 One colour scheme could be blue, with the darker shades closer to Earth because the atmosphere is denser in these layers.

5 The troposphere is 16 km thick while the exosphere is 600 km thick, but both are shown as the same width. The diagram is representational of the layers only, not the distances.

6 All the air that supports life is in the troposphere. Weather elements are also here.

7 No. It is all around Earth. The diagram is drawn this way to save space and to emphasise the atmospheric layers.

8a There is no clear division between one layer and the other.

8b Gradual merging rather than clear lines would be more accurate.

10 True False – the troposphere affects people the most. False – the mesosphere contains no clouds. False – the ozone layer is in the stratosphere. True False – the ozone layer is getting thinner due to human activities. This is causing more UV radiation to reach Earth.

4 Heating places on Earth

1 Sun, wave radiation, heat, clouds, absorb, 6, atmosphere, dust, water, 55, reflected, atmosphere, 23.5, Equator, more, high, same, different, Equator, less, more, Poles, angle, colder, near

2 In summer, Earth's tilt makes a more direct angle of incoming solar radiation with less atmosphere to travel through compared to the other seasons, in particular winter. In winter the angle means that the radiation travels through more atmosphere and is spread out over a much larger area of Earth for the same amount of incoming radiation as during summer.

3 Much of the heat is trapped inside the greenhouse so the temperature is higher than the surrounding air. This is better for plant growth as long as they get enough water.

5 The levels of some greenhouse gases, such as carbon dioxide and methane, are being increased due to human activities thus meaning more heat than normal is trapped in the atmosphere. This 'enhances' the normal greenhouse effect.

6a Higher water vapour, carbon dioxide emissions and methane.

6b Methane is produced by cattle and some plantations such as rise. Both products increase with greater population.

6c Higher carbon dioxide and methane levels.

7 It is a slow process, but overall the concern is that the average global temperature will rise. This is 'global warming'.

5 Cloudy skies above

1 Cloud names **a** – cumulonimbus **b** – nimbostratus **c** – cirrus **d** – cirrocumulus **e** – cirrostratus **f** – altocumulus **g** – altostratus **h** – cumulus **i** – stratocumulus **j** – stratus **k** – fog.

Vertical scale in metres should show 2000 metres at border of low and middle, 600 metres at border of middle and high, and the rest should be correlated to these.

Title should be something like 'Types of Clouds' or 'Cloud Types and Where They are Found'

2 A schematic diagram shows 'the scheme' of things – a system, a plan or a layout. It locates objects in the correct place relative to each other, but doesn't have to be to scale. It gives a strong visual representation. Understanding of clouds is helped by matching outside observations with the diagram.

3 High – cirrus, cirrocumulus, cirrostratus; Middle – altostratus, altocumulus; Low – cumulus, nimbostratus, stratus, stratocumulus; Extending through all levels – cumulonimbus

4 Cumulus – altocumulus, cirrocumulus, stratocumulus, cumulonimbus or cumulus; Stratus – altostratus, cirrostratus, nimbostratus, stratus; Cirrus – cirrostratus, cirrocumulus, cirrus; Nimbus – nimbostratus, cumulonimbus

6 Rain formation

1 Left diagram – 4, 1, 5, 2, 3 Right diagram – 10, 6, 8, 9, 7, 11

2 Left diagram is Convectional Rain and right diagram is Orographic Rain

3 13, 17, 16, 12, 15, 14. Title is Rain from a Cold Front.

5 True False – rain can form if air rises quickly and condenses. False – a rainshadow forms on the leeward side of a mountain. False – cold air is drier and heavier than warm air. False – convectional rain occurs in local conditions that are warm and humid. True.

6 A schematic diagram shows 'the scheme' of things – a system, a plan or a layout. It locates objects in the correct place relative to each other, but doesn't have to be to scale. It gives a strong visual representation. Each of the rain formation diagrams shows a system.

7 A river system

1 delta – 8, estuary – 7, flood plain – 6, lake – 4, meander – 5, mouth – 9, sea – 10, source – 1, tributary – 3, waterfall – 2

3 Delta is the third letter of the Greek alphabet shaped like a triangle. Many deltas are similar to this shape.

4 A town could be located close to the coast and river on the plain, but preferably not directly on the flood plain.

5 natural, water, zones, wettest, year, shallow, water, wetlands, riparian, land, river, natural, upland, drier, wetlands, sponges, wet, vegetation, water, limit, forest, higher, absorb, breeding, fish, insects

8 The moving Earth

1 Divergent plate movement is where two tectonic plates move apart horizontally in opposite directions, leaving a gap in between.

Convergent plate movement is where two tectonic plates move toward each other, with one sliding over the top of the other. Plates could be pushed up or down where they collide.

Transform plate movement is where two tectonic plates slide past each other in opposite directions.

2 A block diagram shows a 3-dimensional section of Earth in a wedge-like representation, usually with different layers. The diagrams simplify the different types of tectonic movements so that the processes can be easily understood.

3 Where two tectonic plates move – either diverging, converging or in a transform plane – a fault often results, which can lead to an earthquake.

4 Volcano or tsunami

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6 Title – Formation of a Tsunami. Many tsunamis are generated by subduction zone earthquakes – for example, earthquakes that occur where an oceanic plate sinks beneath a continental plate. Some of the earthquake's energy is transferred to the ocean above where it travels in all directions in the form of waves. This is the propagation stage. When the waves reach land the size of the waves increases and low coastal areas can be inundated (flooded), often destroying whatever is in the tsunami's path.

7 The 2004 tsunami was due to an underwater earthquake caused by the compression between the Indian and Burmese plates off the coast of northern Sumatra. The earthquake was very large in geographic extent over an estimated 1200 km of fault line, which slipped about 15 m along the subduction zone of the meeting point of the Australian, Indian and Burmese plates.

9 Desert landforms

1 Alluvial fan – g butte – i stony plain (gibber) – b arch and mushroom rock – k longitudinal dune – f playa lake – d star dune – c badlands – e mesa – h barchan dune – a transverse dune – j wadi – l

2 Rainfall in deserts is low but when it falls it is often comes down in torrents and so doesn't soak into the ground. This results in a lot of surface runoff, resulting in rapidly flowing floods – flash floods. Erosion is heavy and deposition occurs elsewhere.

3 Because Pancho used the badlands and lived at a time in Mexico when there were many bandits, he was probably a Mexican 'bushranger'. He hid in the heavily eroded country. 'Badlands' were named after the 'bad' people who took refuge there.

4 Desert landforms in table in order from top matched with description from table in question 1 – transverse dunes (j); longitudinal dunes (f); barchan dune (a); star dune (c); arch and mushroom rocks (k); badlands (e).

Remaining landforms in block diagram (clockwise from top right) are mesa, gibber plain, butte, playa lake, wadi, alluvial fan

10 From natural to built

1 It gives a 3D view, which can also reveal things under the surface, eg the depth of the river and different earth layers.

2 River, forest, kangaroos, birds, wetlands, grassland, Aboriginal camp, Aboriginal men hunting and fishing. The natural environment dominates, with only a small amount built and managed.

3 River, smaller amount of natural forest, farming homesteads with sheep, paddle steamer on river, 2 roads, fences, timber mill on hill slope, township built around a mining operation. It is mainly a built environment with some managed sections, and only remnants of the natural environment.

4 Europeans have displaced the Indigenous group and built houses and other structures related to work, plus things associated with transport. They are extracting mineral resources, logging timber and introducing foreign farm animals. The river is being used as a major transport route.

5 River has been straightened, bridge, marina, townhouses, high-rise buildings, cemetery, quarry, parkland, a larger urban community.

5a Built. **5b** – Changes have occurred due to population growth, technology, greater prosperity.

6 3D view of a section of Earth's surface. **6a** – It is not to scale as it is just a visual representation.

6b – a cutaway section has been added to show more detail of a smaller section – the marina.

7 (i) – a series of 3 maps showing landuse and (ii) a combination of artist impression and photographs of the same area.

8 Student's own answer.

11 Profiling vegetation and soil

1a A profile is a side view summary, which is often sketched. This is a profile of a rainforest.

1b Measurements are the scale, which allows tree heights to be estimated on both the sketch and the photo.

1c Emergents – the tallest trees, around 50 metres, that 'emerge' above the main canopy. Canopy – the main body of high trees at around 30–40 metres, which has a blocking effect. Understorey – more scattered trees and high shrubs from 10–20 metres. Floor – small shrubs and ferns close to the ground.

2 Some labels could be – cutting down forest trees, sheep and cattle grazing, cropping, mining, roads and transport, housing, river pollution

4 Weathering is the chemical breakdown of rock into smaller particles. Leaching is rapidly seeping water that removes minerals and salts as it moves downward. Decomposition is the breakdown of organic matter. Organic matter is derived from living things such as plants, trees, grasses, animals and insects. Humus is well decomposed organic material near the top of the soil profile.

5 There is a lot of organic material (thick leaf litter and humus) near the surface so the roots spread out rather than extend down.

6 Rainforest soils are of a poor quality below the O (organic) horizon due to leaching.

7 When trees are cleared there is less organic material. The topsoil could be washed away by heavy rainfall and the O horizon could disappear. This leads to degraded soils, more erosion and heavily silted rivers.

8 Student's own answer, but could indicate that there are different layers of vegetation, soil horizons and how sensitive the rainforest is to change through human impact.

12 Transects show change

1a It is a straight-line through a city north and south of the CBD.

1b 8 zones – CBD, river, old industry, arts and sports, public housing, parks and gardens, new townhouses, middle suburbs

1c Zero is the scale point below the tallest CBD building. The last scale point to the north is 8, and the last to the south is 6. **1d** – 4.5–6 km

1e – The old industrial area is likely to be developed as warehouse apartments, offices, colleges, and café and entertainment areas. This is because the value of the land close to the city centre is too high for industry, and the pollution and transport issues are too difficult.

2a Elevation (altitude), temperature, vegetation.

2b A deciduous forest is at point X.

2c The temperature at point Y is about 8°C or 9°C, as it would be 7.4°C at 3600 metres.

2d It is not a real place as it depicts a scheme or pattern of how vegetation changes with elevation.

3 It is not of a particular place but shows what changes occur in general. These might vary from place to place. For example, there would be no tropical rainforest at places where the temperature is less than 30°C.

4 As altitude increases the vegetation type changes. For example, at a sea-level temperature of 3°C, vegetation changes from tropical to deciduous to evergreen. Above the treeline at 3600 metres there are few or no trees, just shrubs and grass meadows.

5 The transect diagram of Queensville shows the multiple categories of landform, soil, vegetation, transport, settlement and landuse rather than just one or two variables.

6 True False – the soil at point J is shallow and rocky, and the forest had mainly been cleared for sheep and cattle farming. False – point K would be suitable for activities such as walking or bike riding. False – there is a national park at point L, but there are no roads, just walking tracks. False – it is suitable for fruit and vegetable growing. True

13 Manufacturing flows

1a The input is the combination of X and dot symbols, which the student should draw.

1b Conversion **1c** – The output is the diamond symbol, which the student should draw.

1d By graphics. **1e** – Flow diagram. **1f** – raw materials are inputs and manufactured products are outputs.

2a Liquid and dry ingredients, with minor inputs of flavour/colour and stabiliser being added later

2b It is finished at the Harden stage and the product is ice cream.

2c 8 stages

3 Broken arrows are used to show the extra materials (inputs) to be added later in the process.

4 A flow chart or diagram that uses information boxes and arrows to show the flow from start to finish.

5 Colour, small graphics.

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14 Food chains

1 **a** – True. **b** – False. A tertiary consumer eats a secondary consumer. **c** – False. Plants are at the lowest trophic level. **d** – True. **e** – False. If something is harmed or destroyed the parts of the food chain that depend on it, will also be harmed. **f** – False. Most nutritional energy is available at the bottom of the food chain. **g** – True, apart from the exceptional circumstances of where they are eaten by animals such as tigers, lions, sharks or crocodiles.

2 A food chain is a type of flow diagram because it shows stages in the energy flow from producers to consumers.

3 **Across:** 3 – predator, 5 – grass, 8 – unbalance, 10 – carnivore, 11 – affect, 14 – prey, 16 – energy, 18 – consumes, 21 – pyramid, 22 – secondary, 25 – ecosystem, 27 – dead **Down:** 1 – herbivores, 2 – organism, 4 – all, 6 – living, 7 – tertiary, 9 – hawk, 12 – web, 13 – consumer, 14 – producers, 15 – true, 17 – trophic, 19 – source, 20 – false, 23 – and, 24 – food, 26 - the

15 The water cycle

1

Evaporation – c	Infiltration – f	Groundwater runoff – e
Transpiration – h	Surface runoff – iC	Collection – g
Condensation – a	Precipitation – b	Groundwater storage – d

3 There is no start or finish to the water cycle. It is a continuous process and differs to a manufacturing process, which has a start (input) and and end (output).

4 The diagram is not to scale. It is a visual representation of the cycle, not the relative sizes or distances of things.

5 A – 3, B – 5, C – 6, D – 4, E – 2, F – 1

6 The condition of the atmosphere is an element of the water cycle. Clear air produces clean rain. Polluted air with gases containing acids often results from human activities in factories and transport. Polluted air produces polluted rain – acid rain.

7 natural, troposphere, water cycle, end, cyclic, oceans, evaporates, plants, condensation, precipitation, snow, hail, downhill, rivers, trapped, runoff, continuous, acid, polluted, hundreds, contaminated

16 The poverty cycle

1 A continuous cycle has no real start or end. It is perpetual, unless some aspect of it changes.

2 Poor people living in poverty.

3 The main set joining the conditions shows that poverty is a cycle, where one thing leads to another. The other set shows that poverty causes each of the conditions, and that, also, each condition leads to poverty.

4 Student's own opinion.

6 Examples are: Africa all countries are developing and none are developed; Asia – Philippines, Vietnam, Cambodia and most others (developing) except Japan, Singapore, South Korea (developed); Europe – Albania, Bulgaria, Romania, Serbia–Montenegro (developing) and most remaining countries (developed); North America – Costa Rica, Mexico, Jamaica (developing) and Canada, Bahamas and USA (developed); Pacifc (Oceania) – Fiji, Papua New Guinea, Solomon Islands and most others (developing) except Australia and New Zealand (developed); South America – all are developing, although Argentina and Chile could be borderline.

7 There are no developed countries on the continents of Africa and South America, and very few in Asia and the Pacific. Many people in these areas could live in poverty. (Although this does not mean that people don't live in poverty in developed countries.)

8 The title could be Cycle of Disease or similar. It shows the cycle of the transmission of diarrhoea due to the effect of poor sanitation through mosquitoes breeding and carrying the disease, and/or through humans drinking polluted water. It implies that replacing poor sanitation with good sanitation could break the cycle.

9 They are continuous and have no real start or end. Straight-line flow diagrams start with an input/s and end with an output (product or effect).

10 False. There are poor people trapped in their own poverty cycle within developed countries, eg homeless people, the jobless without skills continually reliant on social welfare.

17 Seasonal cycles

1 Earth is spinning on its axis.

2 Night and day.

3 The seasonal cycle. As Earth orbits the Sun, the angle of the axis causes the amount of radiation heating a unit of land to change. In summer, the Sun's rays are more direct, so the heat lost travelling the atmosphere is less and heat is concentrated in a smaller area. In winter, the angle is less direct, so more is lost travelling through a larger part of the atmosphere and heat is spread over a wider area.

5 It is not to scale. The point is to represent Earth's orbit around the Sun, and the change through different times of the year. It can't be drawn to scale as the Earth's orbit is enormous.

8 Linear timelines usually show activities or events over many months or years from a start point to an end. This circular timeline shows the annual activities that are repeated year after year.

18 Similarities and differences

1 Comparing differences and showing similarities between the world's two polar regions.

2 Below zero temperatures, snow and ice, very windy, icebergs, dark for 6 months each year.

3a – whales, sea birds, seals **3b** – Emperor penguins **3c** – polar bears, wolves, reindeer

4 Antarctica is a separate continent because it is a large piece of land whereas the Arctic is ice surrounded by land, which is part of several continents – Europe, Asia and North America.

5 The Arctic because there are no indigenous people on the land of Antarctica, only research scientists and a few tourists.

6a – Great Beauty Alpine park – 36 in summer and 59 in winter
6b – Rafferty's Lakes Resort – 58 in summer and 55 in winter
6c – Sunshine Pioneer Farm – 69 in summer and 76 in winter **6d** – The figures in small circles show the students who didn't want to go to any of the choices or did not offer a selection. **6e** 115 students **6f** – The teacher needed to reach a conclusion about which was the most popular destination and in which season **6g** – The teacher probably chose Sunshine Pioneer Farm in July **6h** – The diagrams helped the teacher easily record the multiple choice options as indicated by the answers to 6a, 6b and 6c, and therefore arrive at a decision.

19 Mind maps show links

1 The tsunami that occurred on Boxing Day, 2004.

2 It indicates that the 2004 tsunami centred on Southeast Asia devastated natural and built features, and had many impacts. The graphics provide instant visual recognition.

3 The 'what, where, when and why' of the main subject.

4 The impacts – both immediate and longer term.

5 No warning is one action statement, which led to the immediate impacts. There are longer-term impacts in the future, so this is another connector.

20 Sketching human impact

2 The sketch shows the natural environment of Crystal Lake and bordering vegetation to the east, south and west. In the north, the built environment dominates with housing estates, farming land and transport routes.

5 Field sketches, if dated accurately and drawn from approximately the same angle, can be compared over different time periods and any changes identified.

21 Structure of organisations

2 The pattern is a pyramid shape, with one symbol in the top layer and many at the bottom. This is fairly typical of many organisations, with few people at the top of the hierarchy and more below.

3 Organisation chart.

4 1 – G, 2 – L/S, 3 – L/S, 4 – N, 5 – G, 6 – L/S (but with element of N), 7 – L/S, 8 – L/S, 9 – N, 10 – G, 11 – G, 12 – N, 13 – N, 14 – L/S, 15 – G, 16 – L/S

5b – They are geographic or physical areas of the park and there are 5. **5c** – Functional units are groups who perform specific duties in relation to the overall management and there are 13 of these. **5e** – The Law Enforcement function of Operations North. 5f – 6 **5g** – Project Manager because they are also the Deputy Manager

6 The organisation of Australia's 3-tiered political system from one federal government to 8 state/territory governments and multiple local governments. The technique shows multiple layers, indicating that there is more than one LGA in each state/territory.

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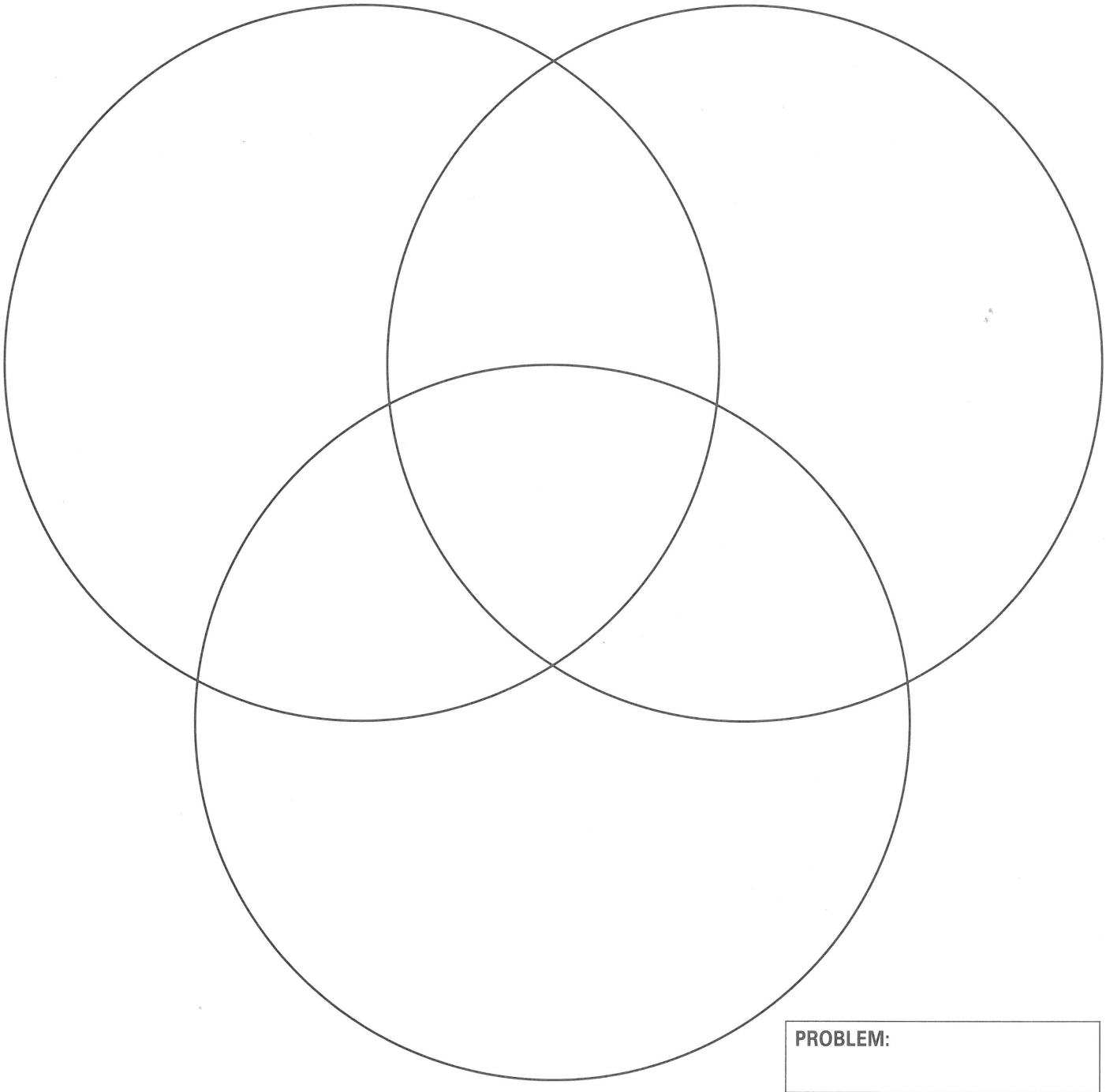
VENN DIAGRAM

Planning template



PROBLEM:

SOLUTION AND EXPLANATION:

PLANNING ELEMENTS:

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